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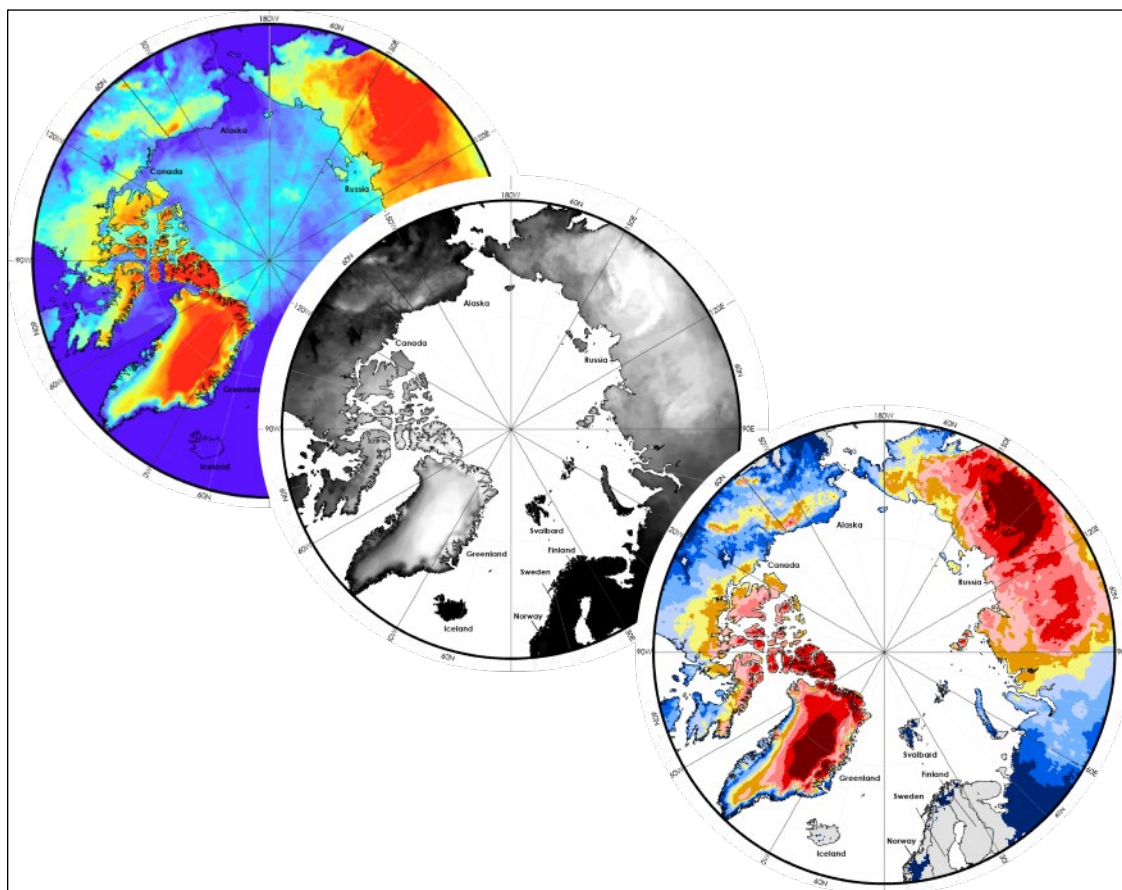
*Environmental Security Technology Certification Program (ESTCP) Alaska
Innovation Landscape Network (AILN)*

Development and Management of Arctic Zonal Characterization Products

Geospatial Database

Evan P. Barrett, Brendon Hoch, Zachary Zody,
Michelle L. Michaels, and Rosa T. Affleck

October 2025



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Development and Management of Arctic Zonal Characterization Products

Geospatial Database

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Abstract

Environmental parameters for operational planning in extreme conditions require accurate knowledge of prevailing meteorological conditions. However, the Arctic region presents unique challenges due to limited observational data and unique geographical conditions. To address the need for such knowledge, this study presents an analysis of Arctic prevailing-conditions using European Center for Medium-Range Weather Forecasting (ECMWF) Reanalysis v5 (ERA5) Data from 1991 to 2020. A custom Python-based framework was developed to process and analyze hourly datasets, identifying zones of extreme events and their frequency across multiple temporal scales. The framework uses ArcPy to automate the generation of nearly 40,000 mapped classifications for land masses 60°N and above. This automated pipeline enables both static and dynamic map generation capabilities for operational planning now and in the future. The resulting dataset provides critical spatial and temporal resolution of Arctic prevailing-conditions, enabling more refined characterization of extreme prevailing-conditions across the circumpolar region.

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Preface

This study was conducted for the Environmental Security Technology Certification Program (ESTCP) under Project NH23-8375, “Arctic Zonal Characterization and Operational Thresholding (AZCOT).” The technical monitor was Mr. Kevin Hiers, ESTCP Program Office, Office of the Deputy Assistant Secretary of Defense (Energy Resilience and Optimization), Washington, DC.

The work was performed by the Terrestrial and Cryospheric Sciences Branch, the Force Projection and Sustainment Branch, and the Engineering Resources Branch of the Research and Engineering Division, US Army Engineer Research and Development Center–Cold Regions Research and Engineering Laboratory (ERDC-CRREL). At the time of publication, Dr. John Weatherly, Dr. Adrian K. Doran, and Dr. Amy E. Cotnoir were branch chiefs, respectively; and Dr. Orian Z. Welling was division chief. The acting deputy director of ERDC-CRREL was Dr. Melisa Nallar, and the director was Dr. Ivan P. Beckman.

This research was supported in part by appointments at the Research Participation Program administered by the Oak Ridge Institute for Science and Education (ORISE), through a cooperative agreement between the US Department of Energy and the US Army ERDC-CRREL.

LTC Joshua M. Haynes was commander of ERDC, and Dr. Beth C. Fleming was the director.

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1 Introduction

The US Army Engineer Research and Development Center–Cold Regions Research and Engineering Laboratory (ERDC-CRREL) is involved in several cold-weather operations, including the development of infrastructure and the planning of missions in extreme weather. This dataset provides information regarding the prevailing conditions across the Arctic as well as details of extreme events, including how extreme they are, how frequently they occur, and how long they occur for within a given day. The final data products are derived from gridded ($0.25^{\circ} \times 0.25^{\circ}$) ECMWF (European Center for Medium-Range Weather Forecasting) ERA5 (ECMWF Reanalysis v5) reanalysis data and are available as GeoTIFFs, ArcGIS Pro Layer Files, or PDFs.

This document details the dataset, including how each file was created, the process for designing the maps, automation techniques in Python, a user interface for creating new maps at custom time intervals, and the storage of the data.

1.1 Dataset Product Name and Keywords

The dataset product is developed as Arctic Zonal Characterization Maps and Data. The specific keywords providing parameter names and definitions are as follows:

- Temperature
- Air temperature
- Surface temperature
- Windchill temperature
- Soil temperature
- Wind
- Wind speed
- Wind direction
- Snow
- Snow depth
- Snow load
- Arctic
- Zonal classification
- Threshold

1.2 Data Temporal Extent

The dataset includes data from 1991 to 2020 during the winter months (October–March), starting with January 1991 and ending with December 2020. The original ERA5 dataset contains hourly data for each meteorological variable, and the observational dataset contains meteorological data recorded between January 1991 and December 2020, although not all stations have the complete 30 years of data.

Finalized maps and their associated files are broken up into six different static timescales shown in Table 1.

Table 1. Timescales for data analysis.

Timescale	Description
Six Hourly	Six-hour periods (00:00UTC–05:59UTC, 06:00UTC–1159UTC, 1200UTC–1759UTC, 1800UTC–2359UTC)
Daily	Daily periods (0000UTC–2359UTC)
Weekly	Weekly (7–8 days) periods
Biweekly	Every two weeks (14–16 days) periods
Monthly	All days within a specified month
Seasonal	1 October–31 March

An example of data at different intervals can be seen in Figure A-1 in the Appendix. In addition to the static timescales above, the user interface allows for dynamic definitions of timescales (Section 6).

1.3 Data Geographic Extent

The ERA5 data were collected for latitudes greater than or equal to 60°N across all longitudes. All meteorological variables have a spatial resolution of $0.25^{\circ} \times 0.25^{\circ}$, with the exception of snow depth (derived from ERA5 land data) which has a spatial resolution of $0.10^{\circ} \times 0.10^{\circ}$.

1.4 Data Types

Data files include NetCDF files (.nc), GeoTIFF files (.tif), ArcGIS Pro layer files (.lyrx), and PDF files (.pdf). Python code can be found as Python files (.py) or Jupyter Notebooks (.ipynb). The user interface directory also contains shapefiles (.shp) and ArcGIS Pro files (.aprx).

1.5 Data Organization

There are three main directories: *Data*, *Atlas*, and *App*. *Data* contains the raw data from which the maps were created. *Atlas* contains static maps and their associated files as created in Sections 3–6. *App* contains everything needed for the user interface, as described in Section 6.

The *Data* directory is broken down as follows:

- The main directories are the 6 months within the study period (*oct*, *nov*, *dec*, *jan*, *feb*, and *mar*).
- Within each month, there are 29–31 subdirectories representing the days within that month. For example, information for 7 November would be in `/data/nov/07/`.
- Within each days folder, there are netCDF files for each meteorological parameter, one for each hour in the study period.
 - The files follow the naming convention of `YYMMDDHH.VARIABLE.nc`. For example, 2 m temperature data from 7 November 2005 at 15:00 UTC would be found at `/data/nov/07/05110715.2T.nc`.¹
 - The meteorological parameters directly from ERA5 are 2 m temperature (2T), skin temperature (SKT), soil temperature (level-1) (STL1), snow water equivalent (SWE), 10 m wind velocity in the eastward direction (10U) and northward direction (10V), and snow depth (SD). Additionally, snow load (SL), windchill temperature (WCT), 10 m wind speed (WS10), and wind direction (WD10) were calculated for each hour, as described in Section 2.2.

The *Atlas* directory is broken down as follows:

- The main directories of the Atlas are the 6 months within the study period (*oct*, *nov*, *dec*, *jan*, *feb*, and *mar*).
- Within each month, there are many subdirectories. These have the general structure as follows:

1. For a full list of the spelled-out forms of the units of measure used in this document and their conversions, please refer to *US Government Publishing Office Style Manual*, 31st ed. (Washington, DC: US Government Publishing Office, 2016), 248–52 and 345–47, <https://www.govinfo.gov/content/pkg/GPO-STYLEMANUAL-2016/pdf/GPO-STYLEMANUAL-2016.pdf>.

- *Time Interval*—after choosing the month, you must choose the time interval of interest from the following options:
 - *Six Hour*—static six-hour intervals for each day. Within the *Six Hour* directory, you must also choose the day of interest within that month.
 - *Daily*—static daily (00:00 UTC–23:59 UTC) intervals. Within the *Daily* directory, you must also choose the day of interest within that month.
 - *Weekly*—static 7–8 day long periods within the month.
 - *Biweekly*—static 14–16 day long periods within the month.
 - *Monthly*—full month’s worth of data.
- *Variable*—after choosing the time interval, you must choose the meteorological variable of interest:
 - *2T*
 - *SKT*
 - *WCT*
 - *STL1*
 - Wind speed (*WSPD*)
 - Wind direction (*WDIR*)
 - *SD*
 - *SL*
- *Metric*—after choosing the variable, you can choose the metric:
 - *Average*—this is the 30-year average (1991–2020) for each of the meteorological variables within the time interval.
 - *Maximum/Minimum*—these maps show the highest recorded or lowest recorded value, respectively for each respective meteorological variable. These were calculated by looking at each time interval and finding the highest/lowest hourly value within that period for each year, and then finding the maximum/minimum amongst the 30 years.
 - *Frequency*—these maps show the frequency of certain extreme events for each of the meteorological variables. For a given time interval, the number of occurrences above/below the threshold within the 30-year period were counted and divided by the total number of hours within that time period.

- *Consecutive*—these maps show the average daily length of extreme events for each of the meteorological variables. For a given multiday period, the average length of extreme events (in hours) was calculated for each day, and then averaged throughout the 30-year period and for the whole week.

Notably, not all subdirectories exist. For example, there are no *Consecutive* subdirectories within the *Six Hour* directory, because that metric was not calculated at a six-hour time scale. Additionally, the *Consecutive* metric does not exist for *STL1*, *SD*, or *SL*, and only the *Average* metric exists for *WDIR*.

The *App* directory is broken down as follows:

- *ArcGIS*—the ArcGIS project files
- *Output*—files created through use of the UI. These follow the naming convention of {DAY1}_{MONTH1}_through_{DAY2}_{MONTH2}_{VARIABLE}_{METRIC}
- *Templates*—layer file templates for file symbology
- *mapUI.py*—the main script needed to run the UI

1.6 File Naming Convention

Most files within the *Atlas* directory follow the naming convention of {MONTH}_{INTERVAL}_{VARIABLE}_{METRIC}.{FILE FORMAT}

- For *Six Hour* intervals, the time interval is formatted as {DAY}_6h_{hour1}through{hour2}
- For *Daily* intervals, the time interval is formatted as {DAY}
- For *Weekly*, *Biweekly*, and *Monthly*, the time interval is formatted as {day1}through{day2}

Quadrant maps (Section 5.3) include the quadrant (BottomLeft, BottomRight, TopLeft, TopRight) at the end of the file name. For example, a quadrant map looking at 2T in Greenland would be named dec_01_2T_average_BottomLeft.pdf.

1.7 Data Volume

An estimated 1.5 TB of data has been created throughout the project.

1.8 Data Restrictions

Personally Identifiable Information (PII) or other information whose distribution may be restricted by law or national security is not included in this dataset.

1.9 Data Storage

The data will be stored on the Alaska Climate Research Center's (ACRC) website, hosted by the University of Alaska, Fairbanks. Brendon Hoch at CRREL will oversee the data transfer. The data will also be stored on the ERDC DoD High Performance Computing system and on physical hard drives at CRREL.

1.10 Points of Contact (POCs)

- Project point of contact (POC): Michelle Michaels (Michelle.L.Michaels@erdc.dren.mil)
- Data POC: Brendon Hoch (Brendon.Hoch@erdc.dren.mil)
- ACRC POC: Martin Stuefer (mstuefer@alaska.edu)

2 Data Acquisition and Preprocessing

The data acquisition process is described in Section 2.1 of the Technical Report, *Arctic Zonal Characterization and Operation Thresholding*, and mirrors parts of this section.² Readers should refer to the technical report for additional details.

2.1 Shapefiles

Shapefiles for the arctic countries and the geographic lines were obtained from Natural Earth (<https://www.naturalearthdata.com/>) in June 2024 for map design and data clipping purposes (Natural Earth 2024b).

2.2 Reanalysis Data

Data were subset for latitudes between 60° to 90° North latitude to cover all Arctic regions. The data were obtained from the ECMWF Data Store (also known as Copernicus) (ECMWF, n.d.-a). The site provides free access to a variety of ECMWF maintained datasets; it allows both manual downloads and a Python application programming interface (API) for more advanced data acquisition. The ERA5 data were initially obtained from ECMWF in gridded binary (GRIB) format. GRIB is a format based on standards established by the WMO and is typically used by operational centers that produce numerical weather prediction (NWP) output. All parameters were downloaded in a single hourly GRIB formatted file. Parameters selected include 2T, SKT, 10U and 10V wind components, SWE, and STL1. Additionally, at the request of stakeholders, snow depth was obtained from a derivative ECMWF dataset known as ERA5-Land (ECMWF, n.d.-b). This dataset provides higher spatial resolution at 0.1° (approximately 9 km) at the expense of noninclusion of data over oceanic and other water surface areas.

Each GRIB file with multiple parameters was split into an individual parameter GRIB file using the open source wgrib software utility available from NOAA (NCEP 2005). On parameter separation, each separated parameter GRIB file was then converted into Network Common Data Format (netCDF) format in an effort to increase compatibility with other data processing applications. The conversion was performed by the open-

2. Affleck, R. T., M. Michaels, B. Hoch, J. Williams, Z. Zody, P. Toebben, and M. Hefner, 2025, *Arctic Zonal Characterization and Operation Thresholding*, ERDC TR-DRAFT, Engineering Research and Development Center, draft report in process.

source National Center for Atmospheric Research (NCAR) Command Language (NCL) software (NCAR 2019a). NCL allows users to perform a variety of statistical and visual analyses of meteorological datasets. The `ncl_convert2nc` tool included within the package allows for conversion of GRIB files to netCDF (NCAR 2019b).

Seven meteorological variables were acquired from ERA5:

- SKT—the temperature at the surface, in kelvins
- 2T—the temperature at a height of 2 m, in kelvins
- STL1—the average temperature 0–7 cm below the surface, in kelvins
- 10U—the easterly wind velocity at a height of 10 m, in meters per second
- 10V—the northerly wind velocity at a height of 10 m, in meters per second
- SD—the depth of snow on the surface, in meters
- SWE—the amount of water stored in snow, in meters

From those variables, four additional variables were calculated,

- WSPD—the wind speed, in meters per second, at a height of 10 m (Earth Observing Laboratory 2014),

$$\text{WSPD} = \sqrt{V^2 + U^2} . \quad (1)$$

- WDIR—the wind direction at a height of 10 m (Earth Observing Laboratory 2014),

$$\text{WDIR} = \left(270^\circ - \arctan 2(V, U) \times \frac{180^\circ}{\pi} \right) \bmod 360^\circ . \quad (2)$$

- WCT—the temperature felt to exposed human skin, in °F, due to the combined effect of air temperature (2T in Fahrenheit) and wind speed (in miles per hour) (Nelson et al. 2002),

$$\text{WCT} = 35.74 + 0.6215(2T) - 35.75(\text{WSPD})^{0.16} + 0.4275(2T)(\text{WSPD})^{0.16} . \quad (3)$$

- Frostbite time (FT)—the time, in minutes, in which frostbite could occur on exposed, dry cheek skin at a given 2T (Fahrenheit) and 10 m wind speed (miles per hour) based on modeling (Nelson et al. 2002),

$$FT = \left(\left(-24.5 \left(\left(0.667 \left(WSPD \times \frac{8}{5} \right) \right) + 4.8 \right) \right) + 2111 \right) \left(-4.8 - \left((2T - 32) \times \frac{5}{9} \right)^{-1.668} \right). \quad (4)$$

- SL—the downward force caused by the weight of accumulated snow or ice in pounds per square feet, estimated here using SWE (inches) data assuming a horizontal angle (Curtis 2007). To calculate the estimated SL at an angle of θ , one would have to multiply by $\cos\theta$,

$$SL \approx SWE \times 5.2. \quad (5)$$

The variables were then converted to the following units as specified by the stakeholders:

- Temperature—degrees Fahrenheit
- Wind speed—knots
- Snow load—pounds per square foot
- Snow depth—inches

2.3 Meteorological Aerodrome Report (METAR) Data

At the recommendation of project stakeholders, additional meteorological point observational data were obtained to provide limited verification of reanalysis data in key locations. These real-time observations are available in the International Civil Aviation Organization (ICAO) standardized Meteorological Aerodrome Report (METAR) format (ICAO 2021). Typically, each station reports meteorological parameters at a minimum of hourly thresholds, often through automated means. Thirty of years of METAR observations were obtained for stations within the area of interest. In some cases, certain stations were not available for the entire period due to budgetary and technological considerations of the individual countries provided the observations.

Once the raw METAR observations were obtained, an initial processing of the data were undertaken using the open-source Python-METAR software suite (Pollard 2018). Data were processed to extract temperature, dew-point, visibility, wind speed and cardinal direction, and current weather observed at time of observation in a comma-separated value (CSV) format.

Additionally, a date-time code was also included which allowed METAR observations to be more easily combined for the purposes of generating time series statistics. Afterwards, the data were parsed using Python and converted to a data frames for further processing.

METAR data were processed and aggregated from 370 stations in high-latitude regions from 1990 to 2020. The database includes measured data from 12 different countries: Belgium, Canada, Denmark, Finland, Great Britain, Greenland, Iceland, Netherlands, Norway, Russia, Sweden, and the United States. The extent of spatial and temporal range and volume of data (49,417,925 processable recordings) required the development of data cleaning and quality control procedures to prep the data for analysis. This quality control process was done using the Python programming language, and the Arctic Zonal Characterization and Operational Thresholding (AZCOT) library used for manipulating METARs is available on request. The goal was to create a repeatable and flexible process for standardizing and down sampling data while preserving quantitative and categorical data for reanalysis.

For this project, the custom Python library was used to generate daily averages of quantitative data and daily modes of qualitative data (e.g., wind direction) for each station across all time periods. The developed functions are repeatable and modifiable for generation of other geographic information system (GIS) friendly data conversions of METAR subsets.

Each METAR observation file was individually cleaned to ensure consistency across stations. Key steps included the following functions in the library are described in Table 2 at the end of this section.

1. Convert decoded METARs to pandas data frames.
2. Geolocate each recording with station location.
3. Parse datetime into numeric format.
4. Convert visibility to numeric format.
5. Split wind into qualitative wind direction and quantitative wind speed values.
6. Restructure data frame for better readability.
7. Down sample to daily averages and modes.
8. Export to GIS readable format.

Table 2. Description of functions in the Arctic Zonal Characterization and Operational Thresholding (AZCOT) library.

Function	Output	Description
<code>stationFilter()</code>	Pandas data frame	Filters the METAR stations based on specified countries or "all" to return all stations.
<code>readMETARs()</code>	List of Pandas data frames	Pulls in METAR files as a list of dataframes based on the desired sorting by station and month.
<code>timeCleaning()</code>	List of Pandas data frames	Processes time data of passed list of dataframes. Splits datetime format into its constituent numeric components and adds those columns to each dataframe in the returned list.
<code>tempCleaning()</code>	List of Pandas data frames	Adds Fahrenheit temperature column to each METAR dataframe passed in the list.
<code>visCleaning()</code>	List of Pandas data frames	Processes visibility data of each METAR dataframe passed in the list. Handles exceptions and converts string fractions to numbers. Overwrites existing visibility value in each dataframe.
<code>windCleaning()</code>	List of Pandas data frames	Processes wind data of each METAR dataframe passed in the list. Splits data into wind speed and wind direction columns. Processes identified cases, but may not capture every possible entry. Adds new columns to each existing dataframe.
<code>structureFrame()</code>	Pandas data frame	Creates new dataframe combining each METAR dataframe passed in the list. Only works if post processing has been completed. Reorders columns, preserves relevant METAR data, and concatenates list into single dataframe for export and use in GIS.
<code>uniqueYears()</code>	Dictionary	Examines each METAR dataframe passed in the list for the years data were recorded. Counts the number of unique years and extracts those years into a dictionary.
<code>sampleScreening()</code>	Pandas data frame	Processes data of all METARs in restructured dataframe. Screens the data range and observation frequency for all stations by year.
<code>averageData()</code>	Pandas data frame	Downsamples data in the master METAR dataframe passed. Computes daily averages of numeric data and finds the daily mode of text data. Restructures into new dataframe that contains only the downsampled entries.
<code>sampleFrequencies()</code>	Pandas data frame	Analyzes observation frequency for METAR stations by year.
<code>top_periods_per_station()</code>	Pandas data frame	Extracts the top <i>N</i> most common period values for each station.

3 Metric Calculations

Metric calculations were performed on hourly netCDF files using the Xarray Python library. The following four main metrics were calculated: the average value, the highest/lowest recorded value, the frequency of extreme events, and the length of those extreme events. In addition to these main metrics, additional extreme and snow metrics were calculated.

3.1 Average

For each pixel, the average value of the meteorological parameter was calculated as an arithmetic mean,

$$A_{i,j} = \frac{1}{n} \sum_{k=1}^n a_{i,j,k}, \quad (6)$$

where

A_{ij} = average at a latitude (i) and longitude (j),
 n = number of hours in time interval, and
 $a_{ij,k}$ = the k th reanalysis value at a latitude i and longitude j (Plate 1).

Plate 1. Pseudocode for Equation (6).

```
ALGORITHM Calculate Average Recorded Value

INPUT Hourly NetCDF files for each meteorological variable
OUTPUT Daily NetCDF files for each meteorological variable

// Main function
FUNCTION CalculateAverage(day)
    fileList ← GET all meteorological files for specified day
    hourlyData ← empty list

    // Load all hourly data for the day
    FOR EACH file IN fileList DO
        data ← READ meteorological data from file
        ADD data TO hourlyData
    END FOR

    // Combine hourly data into a single dataset with one column per hour
    combinedData ← MERGE all hourlyData into single dataset
    results ← empty data frame

    // Calculate average value for each location
    averageValues ← CALCULATE average value across all hours
    results["average_VARIABLE"] ← averageValues
```

3.2 Extreme

For each pixel, the highest or lowest recorded value of the meteorological parameter was calculated. The highest value was found for snow depth, snow load, and wind speed. The lowest value was found for the temperature parameters,

$$E_{ij} = \max\left(\{a_{ij,k}\}_{k=1}^n\right)$$

OR

$$E_{ij} = \min\left(\{a_{ij,k}\}_{k=1}^n\right),$$
(7)

where

E_{ij} = extreme at a latitude i and longitude j ,
 n = number of hours in time period, and
 $a_{ij,k}$ = the k th reanalysis value at a latitude i and longitude j (Plate 2).

Plate 2. Pseudocode for Equation (7).

```

ALGORITHM Calculate Highest/Lowest Recorded Value

INPUT Hourly NetCDF files for each meteorological variable
OUTPUT Daily NetCDF files for each meteorological variable

// Main function
FUNCTION CalculateExtremes(day)
  fileList ← GET all meteorological files for specified day
  hourlyData ← empty list

  // Load all hourly data for the day
  FOR EACH file IN fileList DO
    data ← READ meteorological data from file
    ADD data TO hourlyData
  END FOR

  // Combine hourly data into a single dataset with one column per hour
  combinedData ← MERGE all hourlyData into single dataset

  results ← empty map

  // Calculate minimum/maximum value for each location
  extrValues ← FIND min./max. value across all hours for each location
  results["extreme_VARIABLE"] ← extrValues
  RETURN results
END FUNCTION

```

3.3 Frequency of Extreme Event

For each pixel, the frequency of extreme events was calculated based on a series of defined thresholds derived from US Army Technical Manual 4-33.31 and conversations with stakeholders (Table 3) (HQDA 2017).

Table 3. Threshold values for meteorological variables

Variable	Greater or Less Than	Thresholds
SKT	≤	-75 °F, -70 °F, -65 °F, -60 °F, -55 °F, -50 °F, -45 °F, -40 °F, -35 °F, -30 °F, -25 °F, -20 °F, -15 °F, -10 °F, -5 °F, 0 °F
2T		
WCT		
STL1		0 °F, 5 °F, 10 °F, 15 °F, 20 °F, 25 °F, 30 °F, 35 °F, 40 °F
SD	≥	8 inches, 15 inches, 20 inches, 40 inches
SL		5 lb/ft ² , 10 lb/ft ² , 15 lb/ft ² , 20 lb/ft ² , 25 lb/ft ² , 30 lb/ft ² , 35 lb/ft ² , 40 lb/ft ² , 45 lb/ft ² , 50 lb/ft ²
WSPD		8 knots, 13 knots, 15 knots, 20 knots, 25 knots, 30 knots, 35 knots, 40 knots, 45 knots, 50 knots

We can define a binary array $B(a_{i,j,k})$, where $Ba_{i,j,k} = 1$ if $a_{i,j,k}$ exceeds the threshold and $Ba_{i,j,k} = 0$ otherwise. Then we can define the frequency as

$$F_{i,j} = \frac{1}{n} \sum_{k=1}^n B(a_{i,j,k}), \quad (8)$$

where

- $F_{i,j}$ = frequency at a latitude i and longitude j ,
- n = number of hours in time period,
- $a_{i,j,k}$ = the k th reanalysis value at a latitude i and longitude j , and
- $B(a_{i,j,k})$ = binary function, 1 if $a_{i,j,k}$ exceeds the threshold, else 0 (Plate 3).

Plate 3. Pseudocode for the algorithm in Equation (8).

```

ALGORITHM Calculate Frequency of Extreme Event

INPUT Hourly NetCDF files for each meteorological variable
OUTPUT Daily NetCDF files for each meteorological variable

// Main function to process daily meteorological data
FUNCTION ProcessDay(day)
  fileList ← GET all meteorological files for specified day
  hourlyData ← empty list

  // Load all hourly data for the day
  FOR EACH file IN fileList DO
    data ← READ data from file
    ADD data TO hourlyData
  END FOR

  // Combine hourly data into a single dataset with one column per hour
  combinedData ← MERGE all hourlyData into single dataset
  totalHours ← COUNT of hours in dataset

  results ← empty map

  // Calculate frequency of exceeding threshold
  FOR threshold FROM thresholdList DO
    // Create binary array where 1 indicates exceeding the threshold
    binaryData ← CONVERT combinedData TO binary

    // Count hours below threshold at each location
    countsBelowThreshold ← SUM binaryData across hours

    // Calculate frequency as percentage
    freqBelowThreshold ← countsBelowThreshold / totalHours * 100

    // Store results
    results["frequency_" + threshold] ← freqBelowThreshold
  END FOR

  RETURN results
END FUNCTION

```

3.4 Length of Extreme Event

To quantify how long extreme event lasts when they do occur, the average length of these events within a given day is calculated and then averaged across all thirty years of data.

Let $Ba_{i,j,k} = 1$, where $Ba_{i,j,k}$ is the same binary array defined in Section 3.3, for a number of consecutive hours of length c . If there are m separate events within a day when the threshold is exceeded, then the average length of the events is on a given day is

$$\overline{L_{i,j,y}} = \frac{1}{m_{i,j,y}} / \sum_{k=1}^{m_{i,j,y}} c_{i,j,k,y} , \quad (9)$$

where,

$\overline{L_{i,j,y}}$ = average length at a latitude i and longitude j in the year y ,
 $m_{i,j,y}$ = number of threshold-crossing events at i, j in year y , and
 $c_{i,j,k,y}$ = The length of the k th threshold-crossing event at i, j in year y .

For each year y , the average across all thirty years is then

$$\overline{L_{i,j}} = \frac{1}{30} \sum_{y=1}^{30} \overline{L_{i,j,y}} , \quad (10)$$

for each grid point (i, j) (Plate 4).

Note that in this algorithm, whether the previous day ended with a threshold-crossing event is not factored in. Every day is calculated independently of the surrounding days, meaning that there theoretically could be a multiday event that is simply recorded as 24 hours long for each day, or there could be a 12-hour event that takes place over two calendar days and is thus divided into two shorter events. This is primarily due to processing capabilities.

Plate 4. Pseudocode for Equation (10).

```

ALGORITHM Calculate Length of Extreme Event

INPUT Hourly NetCDF files for each meteorological variable
OUTPUT Daily NetCDF files for each meteorological variable

FUNCTION CalculateAverageLength(binaryArray)
    consecutiveCounts ← empty list
    currentCount ← 0

    FOR EACH value IN binaryArray DO
        IF value = 1 THEN
            currentCount ← currentCount + 1
        ELSE
            IF currentCount > 0 THEN
                ADD currentCount TO consecutiveCounts
            END IF
            currentCount ← 0
        END IF
    END FOR

```

Plate 4 (cont.). Pseudocode for Equation (10).

```

// Handle last sequence if it exists
IF currentCount > 0 THEN
  ADD currentCount TO consecutiveCounts
END IF

IF consecutiveCounts is empty THEN
  RETURN 0
ELSE
  RETURN MEAN(consecutiveCounts)
END IF
END FUNCTION

// Main algorithm that processes data for a single day
// Repeat this for each day from October through March
FUNCTION ProcessDay(day, month)
  combinedResults ← empty dataframe

  // Process each year
  FOR year FROM 1991 TO 2020 DO
    fileList ← GET all hourly wind files for specified day, month, year
    hourlyData ← empty list

    // Load all hourly data for the day
    FOR EACH file IN fileList DO
      data ← READ meteorological data from file
      ADD data TO hourlyData
    END FOR

    results ← empty map

    // Combine hourly data into a single dataset
    combinedData ← MERGE all hourlyData into single dataset

    // Process each threshold
    FOR EACH threshold IN thresholdList DO
      // Create binary array where 1 indicates variable exceeds threshold
      binaryData ← CONVERT combinedData TO binary

      // For each grid square, calculate average consecutive periods
      FOR EACH location IN binaryData DO
        avgConsecutive ← CalculateAverageLength(hourly data)

        results[location]["consecutive_" + threshold] ← avgConsecutive
      END FOR
    END FOR

    IF year is 1991 THEN
      combinedResults ← results
    ELSE
      combinedResults ← combinedResults + results
    END IF
  END FOR

```

Plate 4 (cont.). Pseudocode for Equation (10).

```
// Average the results across all years
combinedResults ← combinedResults / (2020 – 1991 + 1)

// Save results
SAVE combinedResults TO file

END FUNCTION
```

3.5 Aggregating Across Timescales

The average, extreme, and frequency metrics were originally calculated at six-hour timescales, while the consecutive length metric was calculated at daily time scales. To aggregate the average, frequency, and consecutive metric data to a larger timescale, the average of the relevant files was found. To aggregate the extreme metric data, the minimum/maximum across the relevant files was calculated.

3.6 First Occurrence of High Snow Depth and Snow Load

If it is assumed that snow depth and snow load values are relatively nondecreasing throughout the Arctic winter due to consistent below-freezing temperatures, then one can calculate the day within the winter where the snow value crosses a specific threshold for each pixel (Plate 5).

The 1991–2020 averaged values for snow depth and snow load was used. The output maps can be seen in Figure A-3 and Figure A-4.

Plate 5. Pseudocode for the algorithm.

```

ALGORITHM Snow Threshold Crossing Day Identification

INPUT Snow depth/load NetCDF files for Arctic winter
OUTPUT NetCDF file with first day of threshold crossing for each pixel

// Main function
FUNCTION FindThresholdCrossingDay(snowData, threshold)
    firstExceedDay ← array of zeroes

    FOR day ← 1 TO 183 DO
        IF snowData[day] ≥ threshold THEN
            firstExceedDay ← day
            BREAK
        END IF
    END FOR

    RETURN firstExceedDay

END FUNCTION

// Main execution

thresholds ← List of thresholds
FOR EACH threshold IN thresholds DO
    result ← FindThresholdCrossingDay(snowData, threshold)
END FOR

```

3.7 Percentiles

In addition to the highest and lowest recorded values as described in Section 3.2, daily 5th and 95th percentile values were calculated in addition to the lowest and highest recorded values, respectively. The percentile value was calculated using the `pandas.DataFrame.quantile()` function, which calculates the value through a linear interpolation method.

Suppose each grid cell has 30 years of hourly data for each day in the study period. Thus, each grid square has 720 data values for each day, denoted as $a_1, a_2, \dots, a_{720}$. The values can then be sorted ascending order, such that $a_{(1)} \leq a_{(2)} \leq \dots \leq a_{(720)}$. Define the index $i = 720 - 1p$, where p is the percentile of interest (either 0.95 or 0.05 for 95th or 5th, respectively). Then define $k = i = \text{floor}(i)$ and the final value is,

$$Q(p) = x_{(k+1)} + (i - k)(x_{(k+2)} - x_{(k+1)}), \quad (11)$$

for each grid cell (Hyndman and Fan 1996).

The differences between the 95th/5th percentile values and the highest/lowest recorded values was calculated for each day of data and then averaged across the whole October–March period. Table 4 shows the statistics for the whole Arctic based on those seasonal average calculations. Figure A-5 to Figure A-11 show the spatial variability in the differences.

Table 4. Statistics for the difference between 5th/95th percentile values and lowest/highest recorded values for the whole Arctic, October–March.

Variable	Minimum	Maximum	Mean	Standard Deviation
2T (°F)	-12.3979	-2.9524	-7.2263	1.6369
SKT (°F)	-15.1020	-0.6522	-8.8224	2.2796
STL1 (°F)	-25.6389	-0.0013	-6.3050	2.8052
WCT (°F)	-17.5395	-2.0235	-10.3820	2.3366
SL (lb/ft ²)	0	36.5328	3.4462	3.0807
SD (in.)	0	21.99	3.61	4.40
WSPD (knots)	0.2344	16.6231	5.3825	2.4241

Because of computer memory limitations, the percentile values were not calculated beyond the daily timescale. One could aggregate the percentile values in a variety of ways, including taking the highest or lowest value within a time interval or averaging the values; however, the resultant value will likely be inaccurate. With additional processing capabilities, one could calculate the monthly and seasonal percentile values from the original hourly data.

4 Validation with METAR Data

4.1 Methodology

Based on the availability of data within the METAR files, both temperature and wind speed were validated while snow depth and snow load were not due to lack of snow depth/load data within the METAR data.

For each day of available METAR data, the associated reanalysis data were compiled. The reanalysis data were not clipped to ensure that coastal regions were not removed from data. The *Extract Multi Values to Points* (`arcpy.sa.ExtractMultiValuesToPoints`) ArcGIS tool was used to extract reanalysis data values for each of the stations on the days where METAR data were available (Esri, n.d.-c). The resultant multi-index data structure contained the METAR station IDs, the date that the METAR data were collected, the data collected at the station, and the corresponding reanalysis data from the same time frame (Table 5).

Table 5. A sample row showing the Station ID, date of observation, observational data from the Meteorological Aerodrome Report (METAR), and ERA5 2 m temperature and surface temperature values from that same time.

ID	Date	METAR Temp	METAR Wind Speed	ERA5 2 m Temp	ERA5 Wind Speed
CWVH	2001/01/01	-18.325000	12.875000	-14.699349	6.529890

For a given day of data, the average temperature and wind speed across the 24 hours (00:00 UTC–23:59 UTC) and the minimum/maximum temperature and wind speed, respectively, was calculated from both the METAR and ERA5 data. A series of statistical measurements were then calculated to understand how the METAR data differed from the ERA5 data (Table 6).

Table 6. Statistical measurements used to compare METAR data to ERA5 data.

Measurement	Formula
Bias	$B_i = (x_{\text{ERA},i} - x_{\text{METAR},i})$
Mean-Absolute Error (MAE)	$\frac{1}{N} \sum_{i=1}^N x_{\text{ERA},i} - x_{\text{METAR},i} $
Mean-Square Error (MSE)	$\frac{1}{N} \sum_{i=1}^N (x_{\text{ERA},i} - x_{\text{METAR},i})^2$

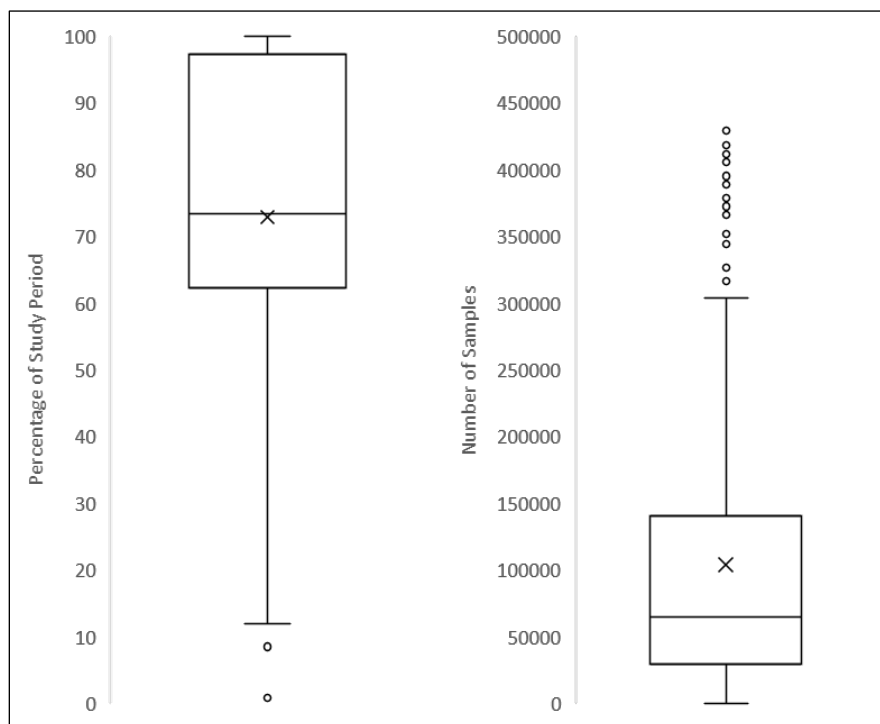
Table 6 (cont.). Statistical measurements used to compare METAR data to ERA5 data.

Measurement	Formula
Root-Mean-Square Error (RMSE)	$\sqrt{\frac{1}{N} \sum_{i=1}^N (x_{\text{ERA},i} - x_{\text{METAR},i})^2}$
Standard Deviation (STD) Ratio	$\sigma_{\text{ERA}} / \sigma_{\text{METAR}}$
Spearman Correlation Coefficient	$1 - \frac{6 \sum_{i=1}^N (x_{\text{ERA},i} - x_{\text{METAR},i})^2}{N(N^2 - 1)}$

4.2 Temporal Analysis.

For each METAR station, the first measurement date, final measurement date, and the total number of samples was obtained. The percentage of the study period captured by each station was calculated by finding the number of days between the first and final measurement date and dividing that by the total number of days within the January 1991–December 2020 study period. The average temporal coverage of the 366 METAR stations was 71.75%, with a 75th percentile value of 97.44%. The maximum number of observations was 429,530, and the average was 1,02555.94 (Figure 1).

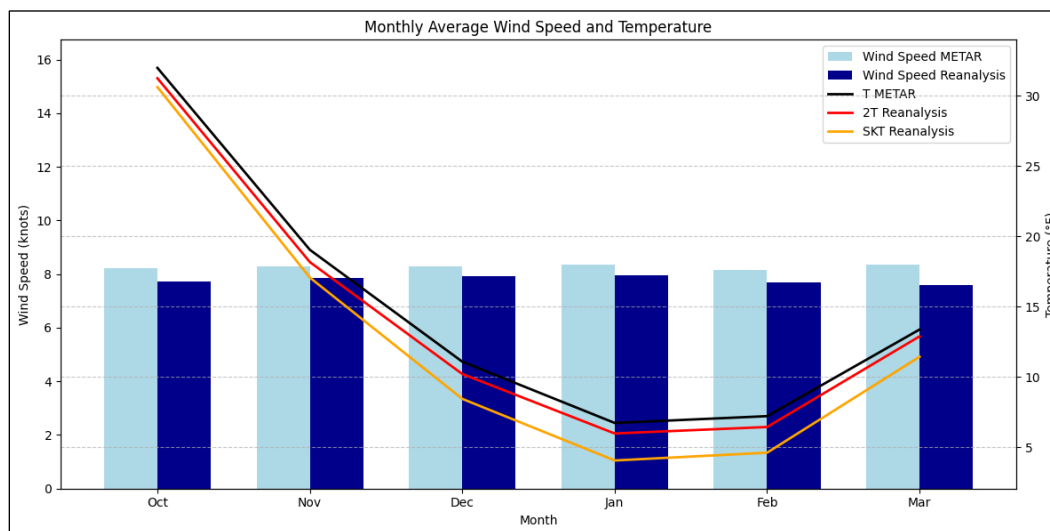
Figure 1. The distribution of temporal resolutions of the Arctic Meteorological Aerodrome Report (METAR) stations.



4.3 Results

The METAR average temperature data correlated strongly with both 2 m temperature ($r = 0.9783$) and skin temperature ($r = 0.9404$), with respective biases of -0.7467°F and -2.1674°F (Table 7). This is expected, as the METAR observations are likely occurring at a height of 2 m and the surface temperature is expected to be colder. The correlation coefficient between the METAR and ERA5 wind speed was $r = 0.7414$ with a bias of -0.4945 knots, showing less than a correlation than the temperature variables but still moderately strong (Table 7). Figure 2 shows the monthly average of the average wind speed and temperature as recorded by METAR and ERA5 data.

Figure 2. Monthly averages of average wind speed and temperature, averaged across the whole Arctic, from METAR and European Center for Medium-Range Weather Forecasting Reanalysis v5 (ERA5) data.



The minimum temperature had slightly weaker but still strong correlations with both ERA5 2 m temperature ($r = 0.9661$) and skin temperature ($r = 0.9237$) (Table 7). The maximum wind speed had a correlation of $r = 0.6991$ with a bias of -3.1885 , indicating that the ERA5 wind speed data skewed lower than the METAR observational data (Table 7). Figure 3 shows the average of the extreme (minimum/maximum) values within the month, and Figure 4 shows the 95th percentile of the extreme values. Based on these two figures, the ERA5 data seems to capture temperature extremes effectively, while there is a slightly larger discrepancy with maximum wind speeds.

Figure 3. Monthly averages of the highest wind speed and lowest recorded temperature, averaged across the whole Arctic, from METAR and ERA5 data.

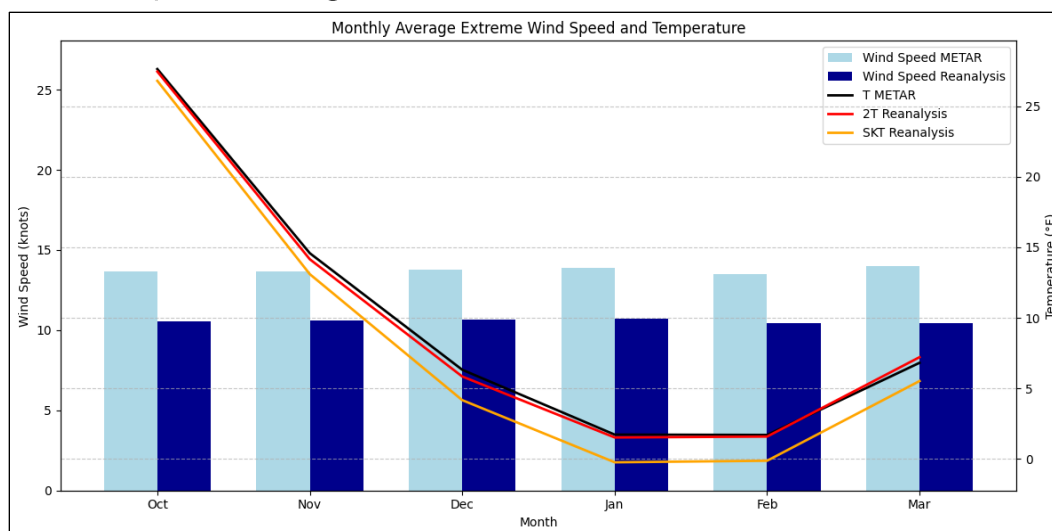
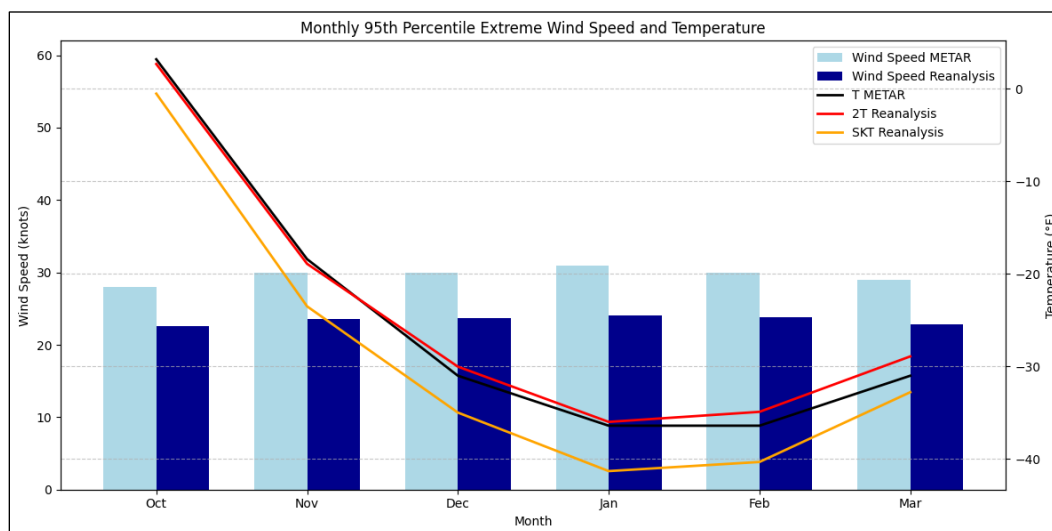


Figure 4. Monthly 5th/95th percentile of the highest wind speed and lowest recorded temperature, from METAR and ERA5 data across the whole Arctic.



Overall, the reanalysis data are relatively consistent with the observational wind and temperature data. Temperature variables demonstrate strong correlations, with modest negative biases and reasonable error metrics, indicating reliable representation of temperature patterns in the reanalysis product (Table 7).

However, wind speed variables show more significant discrepancies, particularly for maximum wind speeds where the correlation drops to approximately 0.70, accompanied by significant negative bias (-3.19) and markedly elevated error metrics (RMSE of 14.21). The notably low

standard deviation ratio (0.42) for maximum wind speed suggests ERA5 slightly underestimates wind speed variability but is still feasible for observing wind trends (Table 7).

These findings indicate that the ERA5 reanalysis data can be used with confidence for temperature-based applications and wind speed analyses. The observed METAR wind speeds trend higher than those from the ERA5 data, likely due to the observational data's ability to capture localized and short-lived wind extremes more effectively than the spatially and temporally averaged ERA5 data.

Table 7. Reanalysis data versus METAR station data statistics.

ERA5 Variable	Metric	Bias	Correlation	MAE	MSE	RMSE	STD Ratio
2T	Average	-0.7467	0.9783	3.2342	22.3673	4.7294	0.9839
	Minimum	-0.1691	0.9661	4.2366	35.6037	5.9669	0.9774
SKT	Average	-0.1691	0.9661	4.2366	35.6037	5.9669	0.9774
	Minimum	-2.1674	0.9404	5.5494	59.6453	7.723	1.0368
WSPD	Average	-1.5721	0.9237	6.6894	81.6559	9.0364	1.0502
	Maximum	-0.4945	0.7414	2.8761	20.9996	4.5825	0.7573

5 Map Creation

5.1 GeoTIFF Creation

NetCDF files produced as the result of the metric calculations in Section 3 were converted to GeoTIFFs using the ArcGIS *Make NetCDF Raster Layer* tool (`arcpy.md.MakeNetCDFRasterLayer`) (Esri, n.d.-d). The GeoTIFFs were clipped to the continental land masses using the *Extract by Mask* (`arcpy.sa.ExtractByMask`) tool with the Arctic countries shapefile (Section 3.1) and an analysis extent of -180° – 180° longitude, 60° N– 90° N latitude (Esri, n.d.-b).

5.2 Symbolology

The symbolology was developed based on iterative feedback from the stakeholders to be both accessible and informative. For temperature, warm colors were chosen to represent warm temperatures and cool colors represent colder temperatures. For wind and snow, warm colors represent dangerous conditions while cool colors represent more moderate conditions. Layer files were created from GeoTIFF files and a series of template layer files using the *Apply Symbolology from Layer* tool (`arcpy.management.ApplySymbolologyFromLayer`) (Esri, n.d.-a).

If one is interested in updating the symbolology, they can do so as follows:

1. Import a GeoTIFF to ArcGIS Pro
2. Right click the layer in the Contents Pane and click “Symbolology”
3. Set the symbolology
4. When satisfied, right click the layer and click “Sharing” and then “Save as Layer File”
5. Save the layer file somewhere you can easily access it
6. Apply the symbolology to any additional GeoTIFFs using the *Apply Symbolology from Layer* tool

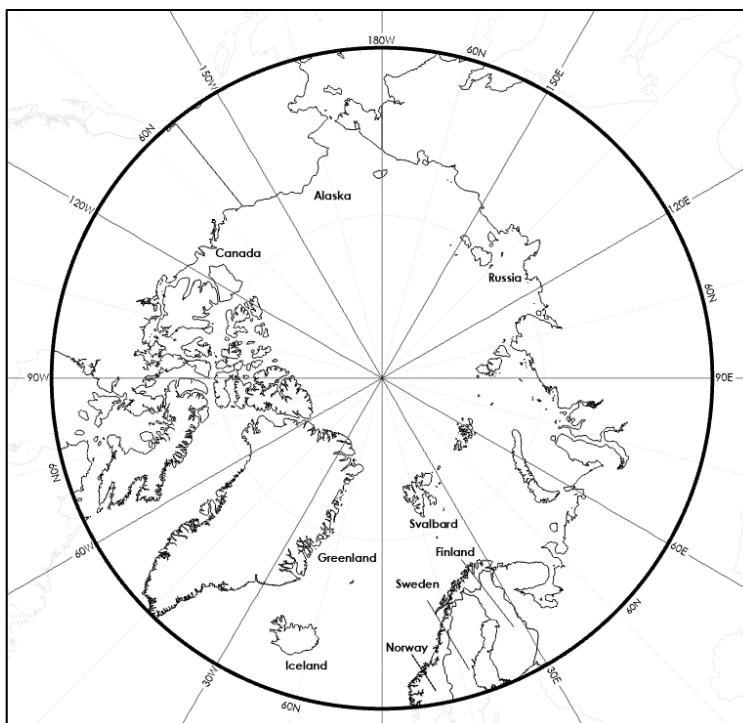
5.3 Map Layout and Design

The map design was developed based on iterative feedback from the stakeholders. The map uses the World Geodetic System (WGS) 1984 Arctic Polar Stereographic projected coordinate reference system, focusing on 60° N and above (EPSG, n.d.; Esri 2025). Arctic countries and Svalbard are labeled and outlined. The 60° N latitude line is emphasized, and the 75° N

latitude line is present in the background. Primary longitude lines occur in 30° intervals, and secondary lines occur in 15° intervals. Map layouts feature a descriptive title, a short subtitle detailing how the data were produced, and a legend.

Based on feedback from project stakeholders, four quadrant maps—in addition to the map of the entire Arctic—were created for each metric and time interval combination. These maps focus on the following areas: 0°E – 90°W (*bottom left* of Figure 5), 90°W – 180°W (*top left*), 180°W – 90°E (*top right*), 90°E – 0°E (*bottom right*).

Figure 5. A blank map showing the design features in the study area.



5.4 Automated Creation

A Jupyter notebook was created to manage the map creation processes (mapCreation.ipynb). Within this notebook there are 119 code cells, broken down by metric, time interval, and variable. The code is dependent on the following Python libraries:

- os
- arcpy
- arcpy.sa
- glob

Each code cell will find all the respective ArcGIS layer files (.lyrx) in that metric, time interval, and variable combination through the glob library. Then, for each file, the code determines the title and subtitle text from the filename and changes them within the map layout accordingly. Then, the layer file is added to the layout, moved to the back to ensure that the design elements are on top of it, exported as a PDF, and then removed (Plate 6).

Plate 6. Pseudocode.

FUNCTION createMap():

INPUT: a ArcGIS layer file

OUTPUT: a PDF map

```
project ← arcpy.mp.ArcGISProject(project_path)
layout ← project.listLayouts("frequencySKT")[0]
map_frame ← layout.listElements("MAPFRAME_ELEMENT")[0]
map ← map_frame.map
```

FOR EACH file **IN** files:

Extract month, day, and threshold (if applicable) from filename

TRY:

```
if element.name = "TitleText":
    element.text ← titleText
if element.name = "TitleText":
    element.text ← subheadingText # update TEXT_ELEMENT
```

```
temp_file ← arcpy.mp.LayerFile(temp_path)
map.addLayer(temp_file) # add the Layer File to map
temp_layer ← map.listLayers()[0] # obtain new layer
map.moveLayer() # moves layer to back
layout.exportToPDF() # export as PDF
map.removeLayer(temp_layer) # remove layer
```

EXCEPT:

Print error and continue

END FOR
END FUNCTION

6 User Interface (UI)

A user interface (UI) was developed for use within Jupyter notebooks to allow the user to generate specific maps of interest, including time intervals spanning two unique days rather than the predetermined time intervals. Additionally, the UI allows the user to generate a contour line at any value. The dependencies and steps for using this UI are located below, as well as the documentation for the existing code if one wishes to edit it or adapt it to another language or GIS software. Notably, the code requires the use of ArcGIS Pro (arcpy).

6.1 File Dependencies

All relevant files should be located in the *App* or *Atlas* directory.

- */App/ArcGIS/mapUI.aprx*—this is the ArcGIS map file used for the map creation process.
- */App/ArcGIS/Design/**—these are the design shapefiles used for the map creation process.
- */App/Templates/**—these are layer file templates used to update the symbology of the GeoTIFFs.
- */App/Output*—this subdirectory is necessary for the output files.
- */App/mapUI.py*—this is the Python script that hosts the relevant functions for the UI to function.
- */Atlas/*/Daily/**/*—the *Atlas* folder contains the daily GeoTIFF files for each meteorological variable.

6.2 Python Dependencies

The following Python libraries or functions are required to operate the UI:

- Python ≥ 3.9
- Jupyter notebook
- fitz
- datetime
- IPython.display
- ipywidgets
- os
- numpy
- glob
- arcpy
- arcpy.sa

6.3 Running the UI

Step 1: Import libraries in a Jupyter Notebook (Plate 7).

Plate 7. Step 1.

```
import mapUI # mapUI.py file
import ipywidgets as widgets
from IPython.display import *
import datetime
import fitz
import os
import numpy as np
import glob
import arcpy
from arcpy.sa import *

arcpy.CheckOutExtension("Spatial") # Spatial Analyst tools
arcpy.env.overwriteOutput = True
```

Step 2: Create and display the UI (Plate 8 and Figure 6).

Plate 8. Step 2.

```
ui = mapUI.createMappingUI()
display(ui)
```

Figure 6. The map creation of the user interface (UI) on generation.

Sliding Date Range Selection

First Month: First Day:

Second Month: Second Day:

1 Variable:

2 Metric:

Threshold (inches):

☐ Contour Line

Contour (inches)

Step 3: Choose your dates, variable, metric, threshold, and other parameters (Figure 7).

Figure 7. The map creation UI with some parameters selected.

The image shows a web-based form titled "Sliding Date Range Selection". It contains several input fields and a button. On the left, there are two rows for date selection: "First Month:" with a dropdown menu showing "January", and "Second Month:" with a dropdown menu showing "February". To the right of these are two sliders for "First Day:" and "Second Day:". The "First Day:" slider is set to 22, and the "Second Day:" slider is set to 21. To the right of the sliders are two more dropdown menus: "Variable:" showing "SKT" and "Metric:" showing "frequency". Below these is a "Threshold (*F):" dropdown menu showing "-65". There is a checkbox labeled "Contour Line" which is currently unchecked. Below the checkbox is a text input field for "Contour (*F):" with the value "0". At the bottom left of the form is a green button labeled "Create Map".

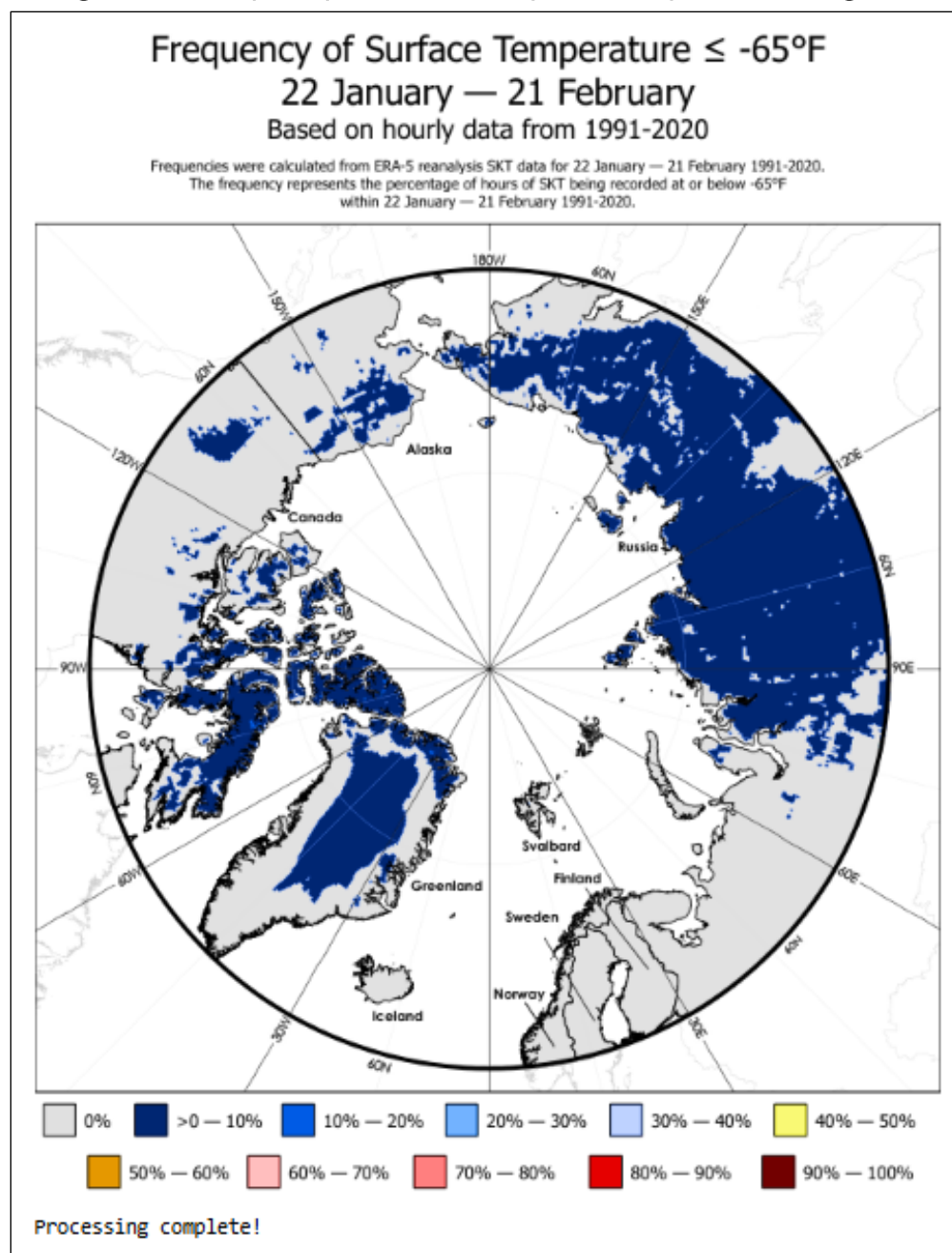
First Month:	January	First Day:	22	Variable:	SKT
Second Month:	February	Second Day:	21	Metric:	frequency
				Threshold (*F):	-65
				☐ Contour Line	
				Contour (*F):	0

Create Map

Step 4: Hit the *Create Map* button and wait.

Step 5: View the resultant map and access files in /App/Output/ (Figure 8).

Figure 8. An example map created from the parameter inputs shown in Figure 6.



6.4 MapUI.py Documentation

MapUI.py is composed of several defined functions that allow it to operate. Below is a detailed look at each of these functions and how they contribute to the overall operation of the program.

6.4.1 UI Creation

- `createMappingUI()` –Main entry point that creates the UI
 - Calls `createDateWidgets()` to set up UI components
 - Contains event handlers (`updateThreshold_visibility`, `updateFirstDayMax`, etc.)
 - Contains `buttonClicked()` which orchestrates the map creation workflow
- `createDateWidgets()` –Creates all UI widgets
 - Depends on `getMaxDays()` to determine the maximum days in a month

6.4.2 Data Processing

- `buttonClicked()` (inside `createMappingUI()`) starts the workflow
 - Calls `isValidDateRange()` to validate dates
 - Calls `generateFilePaths()` to create file paths for data
 - Calls `processTIFFs()` to process GIS data
 - Conditionally calls `getCircularMean()` for wind data
 - Conditionally calls `createContour()` for contour lines
 - Calls `createLyrx()` to create layer files
 - Calls `createMap()` to generate the final map
 - Calls `printPDF()` to display the result
- `isValidDateRange(firstMonth, firstDay, secondMonth, secondDay)` –Validates the selected date range
- `generateFilePaths(firstMonth, firstDay, secondMonth, secondDay)` –Generates paths to data files
 - Depends on `normalizeMonth()`
- `processTIFFs(filePaths, metric, variable, firstMonth, firstDay, secondMonth, secondDay, **kwargs)` –Process GIS data
 - Calls `generateOutputFilename()`
 - Calls `createDirectory()`

- `getCircularMean(filePaths, firstMonth, firstDay, secondMonth, secondDay)` –For wind direction averaging
 - Calls `generateOutputFilename()`
 - Calls `circularMean()`
- `createContour(tif, contourValue)` –Creates contour lines from TIF data
- `createLyrx(tif, metric, variable)` –Applies symbology to layers
- `createMap(lyrx, metric, variable, firstMonth, firstDay, secondMonth, secondDay, **kwargs)` –Creates the final map
 - Depends on various mapping configurations and data

6.4.3 Helper Functions

- `generateOutputFilename(metric, variable, firstMonth, firstDay, secondMonth, secondDay, **kwargs)` –Creates output filenames
 - Uses `monthMap` dictionary
- `createDirectory(filePath)` –Creates output directories if they do not exist
- `circularMean(wdir_paths, wdirPath)` –Calculates circular mean for wind direction
- `printPDF(pdf)` –Converts and displays PDF output
- `getMaxDays(month)` –Returns the number of days in a given month

6.4.4 Dependency Tree

- `createMappingUI`
 - `createDateWidgets`
 - `getMaxDays`
 - `updateThresholdVisibility`
 - `updateContourVisibility`
 - `updateContourVis`

- `updateFirstDayMax`
 - `getMaxDays`
- `updateSecondDayMax`
 - `getMaxDays`
- `updateThresholdValues`
- `updateContourUnits`
- `buttonClicked`
 - `isValidDateRange`
 - `generateFilePaths`
 - `processTIFFs`
 - `generateOutputFilename`
 - `createDirectory`
 - `getCircularMean`
 - `generateOutputFilename`
 - `circularMean`
 - `createContour`
 - `createLyrx`
 - `createMap`
 - `printPDF`

6.4.5 Data Flow

1. The UI is created with `createMappingUI()` which sets up all widgets and event handlers.
2. When the user clicks *Create Map*, `buttonClicked()` is triggered.
3. The inputted date range is validated with `isValidDateRange()`.
4. File paths are generated with `generateFilePaths()`.
5. Data are processed with `processTIFFs()`.
6. If working with wind data averages, `getCircularMean()` calculates wind direction averages.
7. If contour lines are requested, `createContour()` generates them.
8. Map symbology is applied with `createLyrx()`.
9. The final map is created with `createMap()`.
10. The output is displayed using `printPDF()`.

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Appendix: Content

A.1 Example Maps

Figure A-1. The average 2 m temperature based on data from (A) 1 January 00:00–05:59 UTC (six-hour long interval), (B) 1 January 00:00–23:59 UTC (daily interval), (C) 1–15 January (biweekly interval), and (D) 1–31 January (monthly interval).

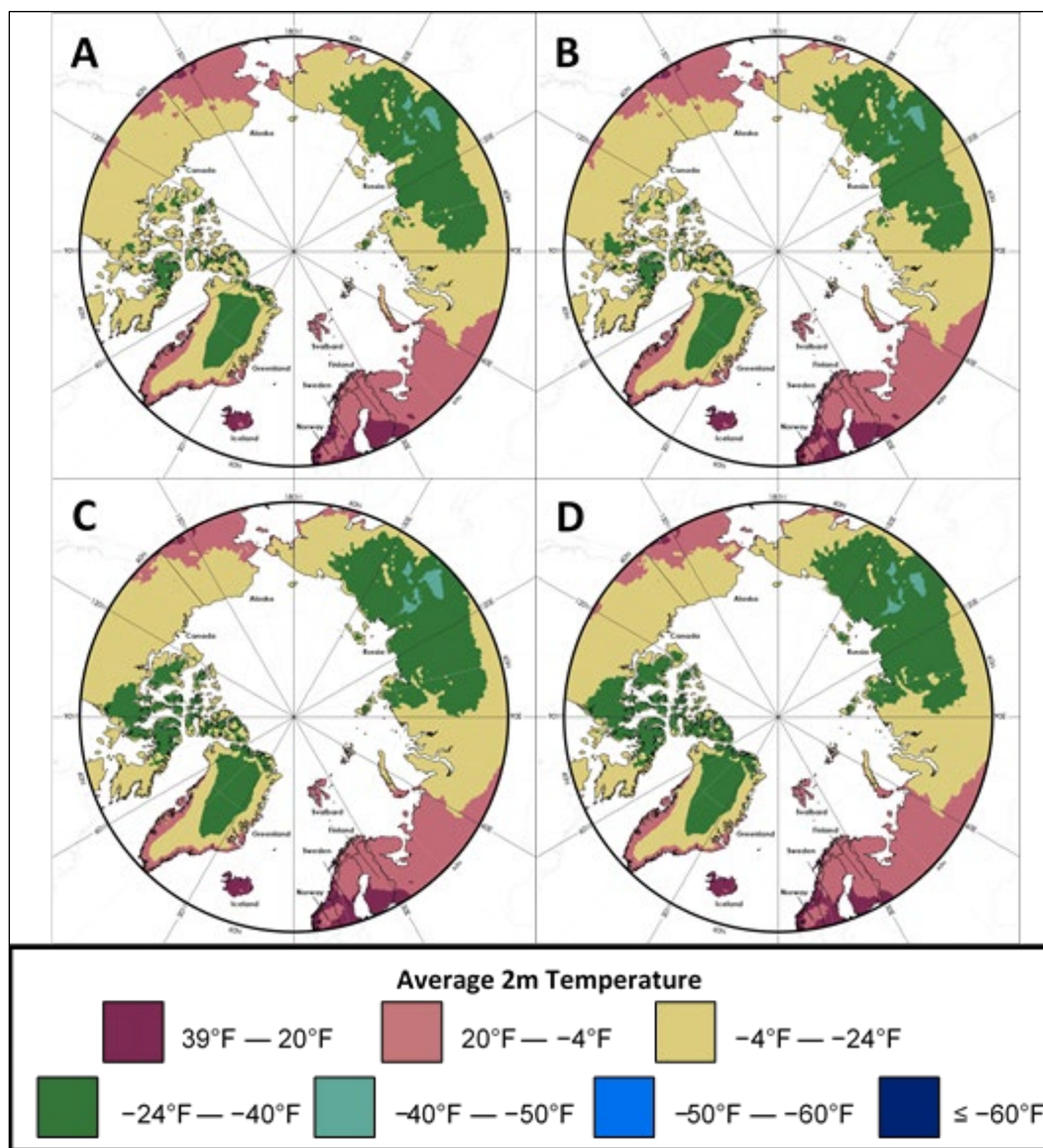
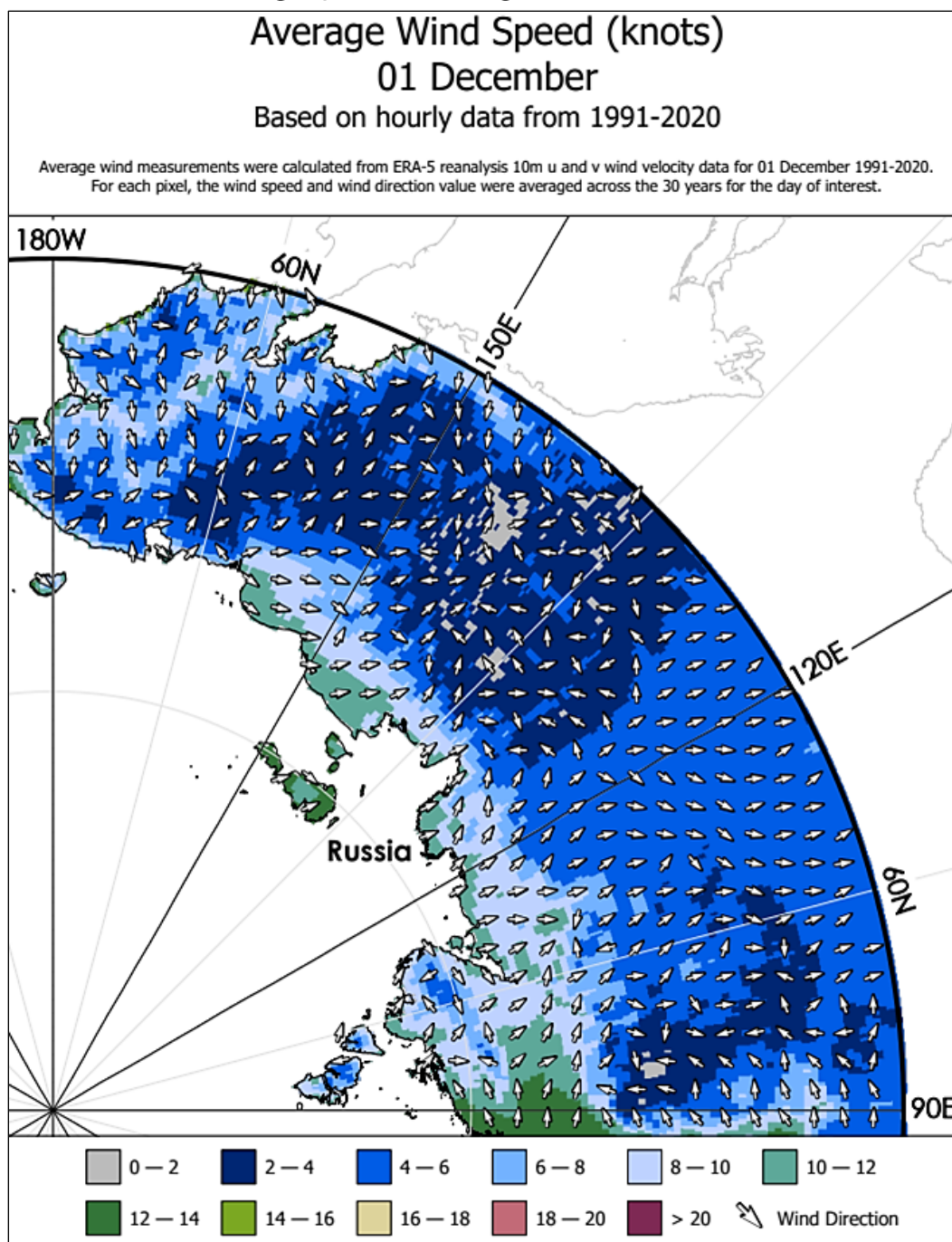


Figure A-2. Quadrant map showing the average wind speed and wind direction in the *upper right* quadrant of the region on 1 December.



A.2 Snow Load/Depth Threshold-Crossing Occurrences

Figure A-3. The amount of time before the first occurrence of snow depth values of (A) 8 in., (B) 15 in., (C) 20 in., and (D) 40 in., based on average 1991–2020 data.

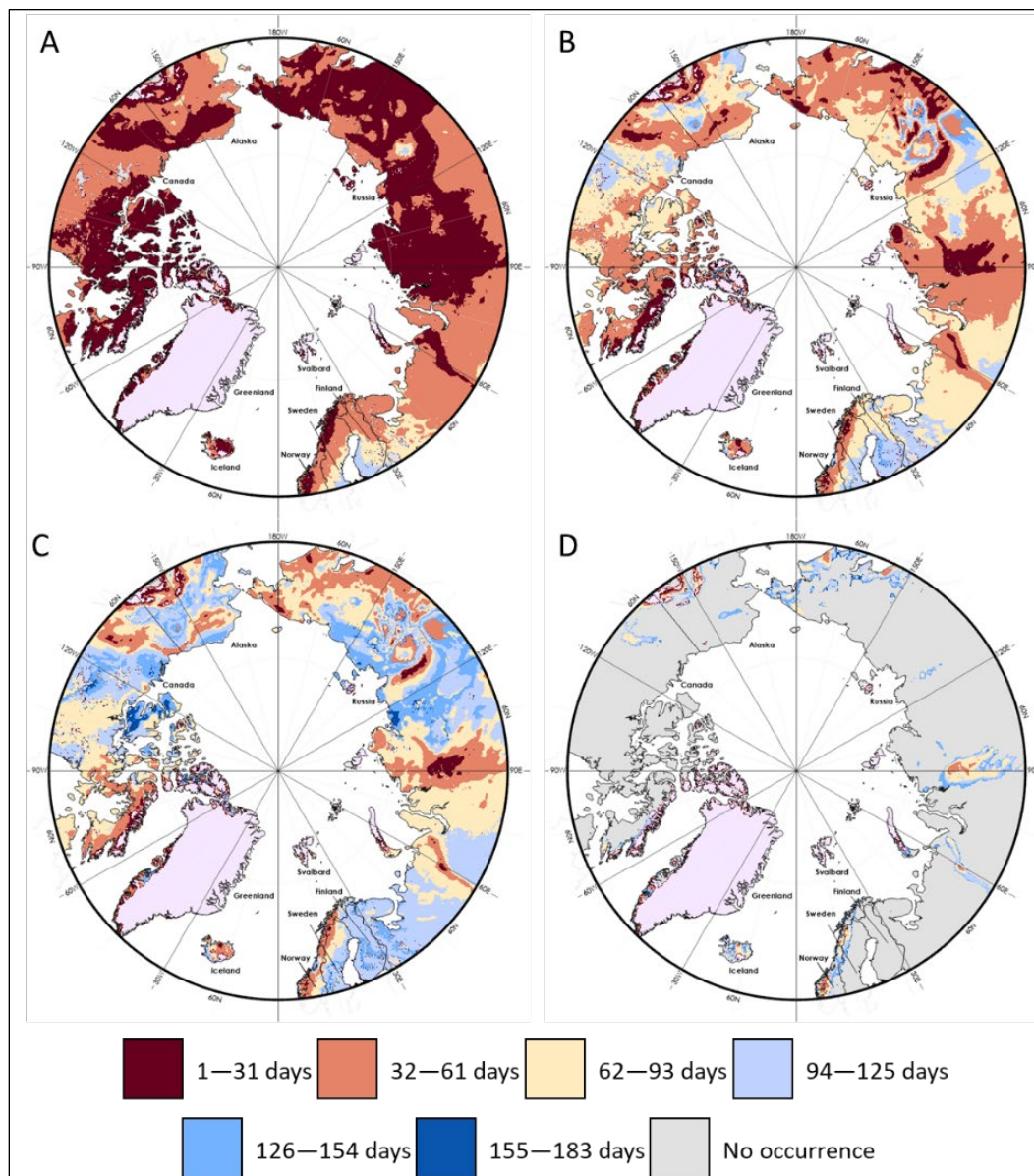
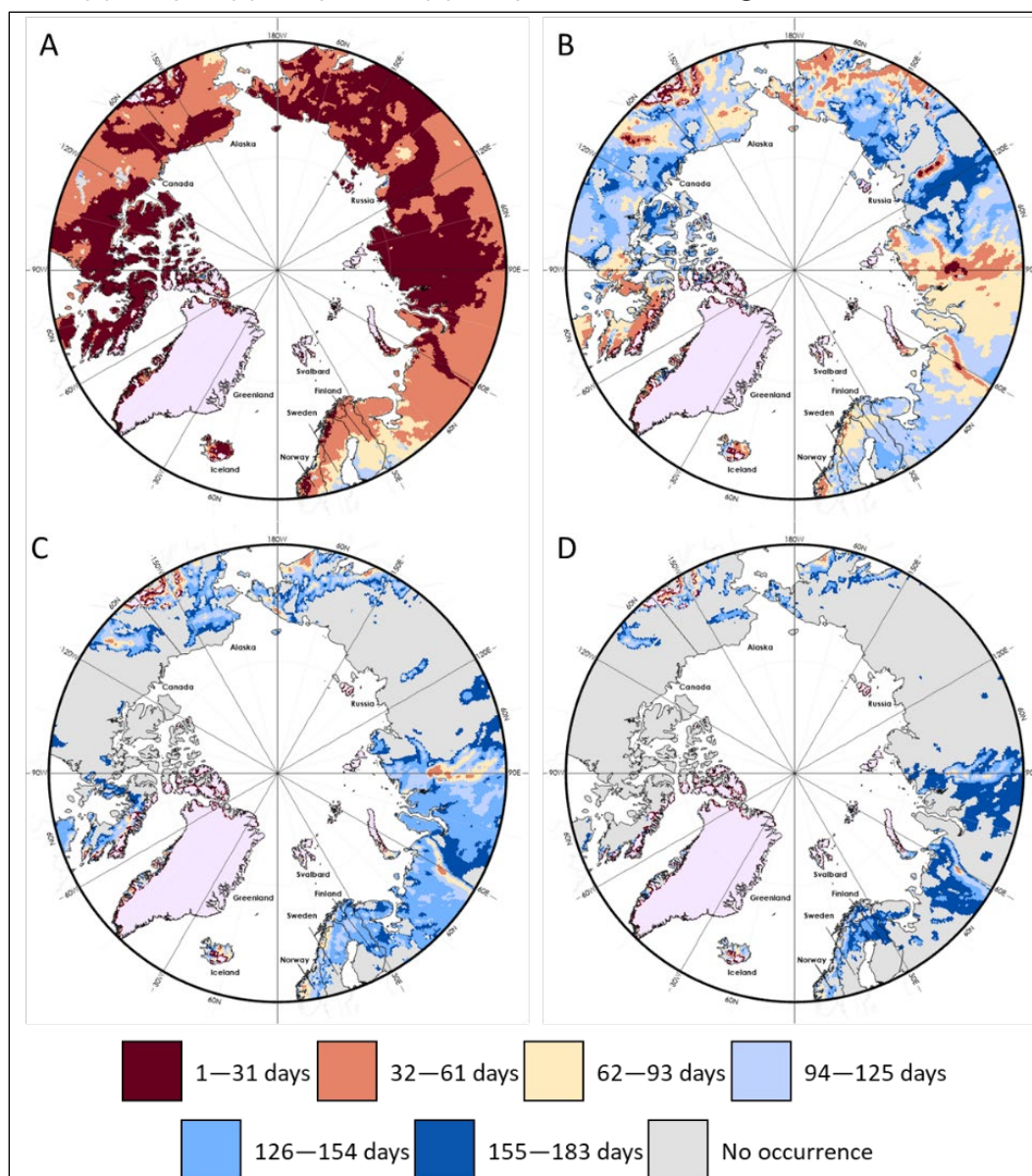


Figure A-4. The amount of time before the first occurrence of snow load values of (A) 5 lb/ft², (B) 15 lb/ft², (C) 25 lb/ft², and (D) 35 lb/ft², based on average 1991–2020 data.



A.3 Percentiles

Figure A-5. The difference between the lowest 2 m temperature (Fahrenheit) and the 5th percentile 2 m temperature, calculated based on daily averaged data computed from hourly 1991–2020 ERA5 data and then averaged across the whole October–March period.

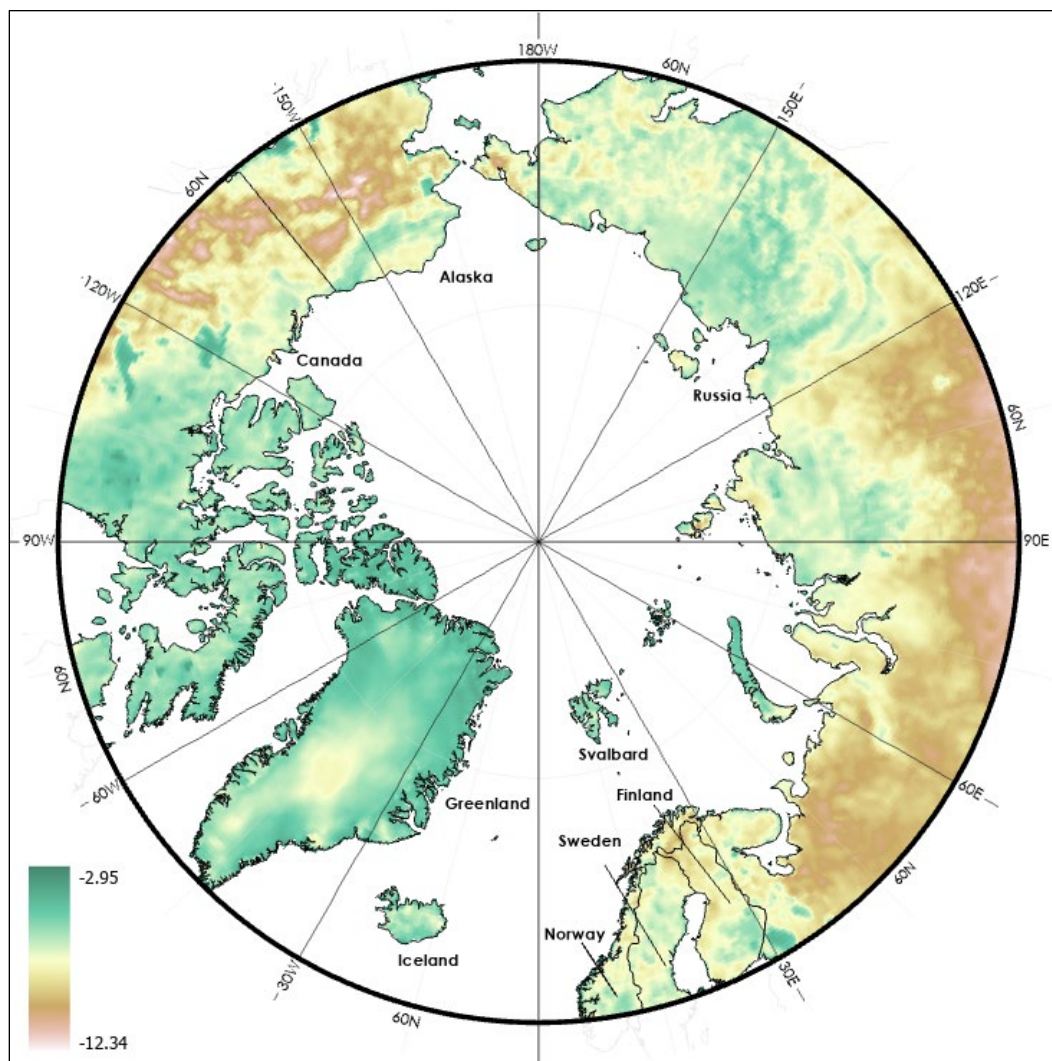


Figure A-6. The difference between the highest snow load value (pounds per square foot) and the 95th percentile snow load value, calculated based on daily averaged data computed from hourly 1991–2020 ERA5 data and then averaged across the whole October–March period.

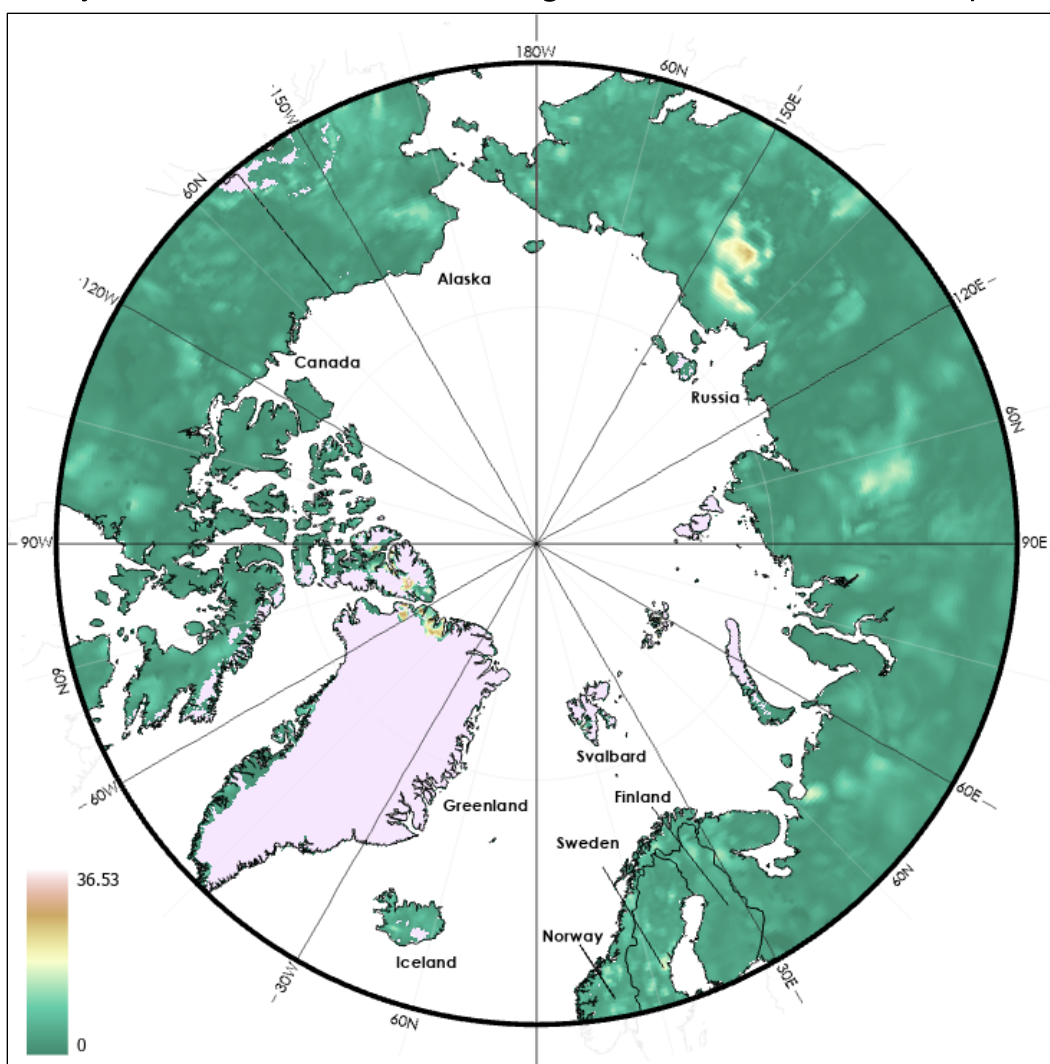


Figure A-7. The difference between the lowest surface temperature (Fahrenheit) and the 5th percentile surface temperature, calculated based on daily averaged data computed from hourly 1991–2020 ERA5 data and then averaged across the whole October–March period.

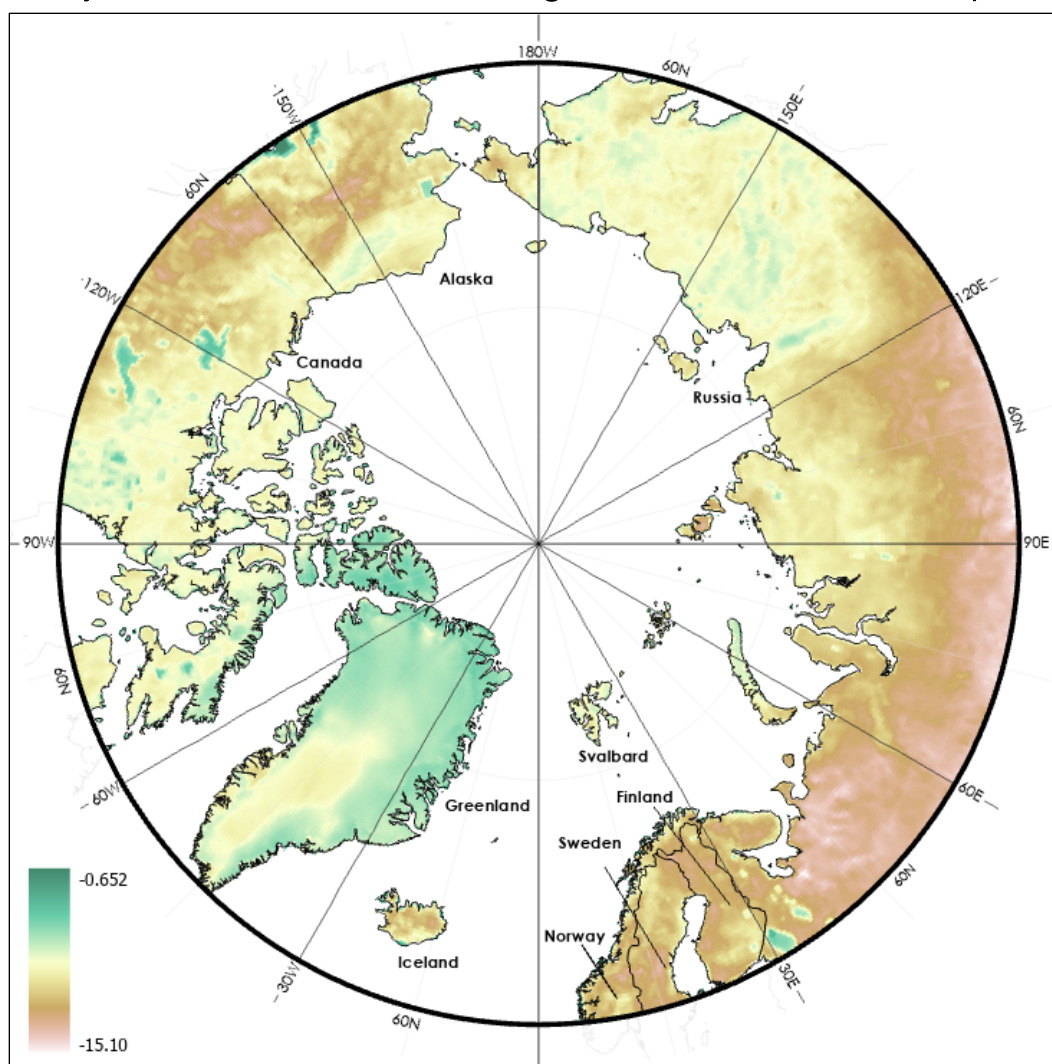


Figure A-8. The difference between the highest snow depth value (inches) and the 95th percentile snow depth, calculated based on daily averaged data computed from hourly 1991–2020 ERA5 data and then averaged across the whole October–March period.

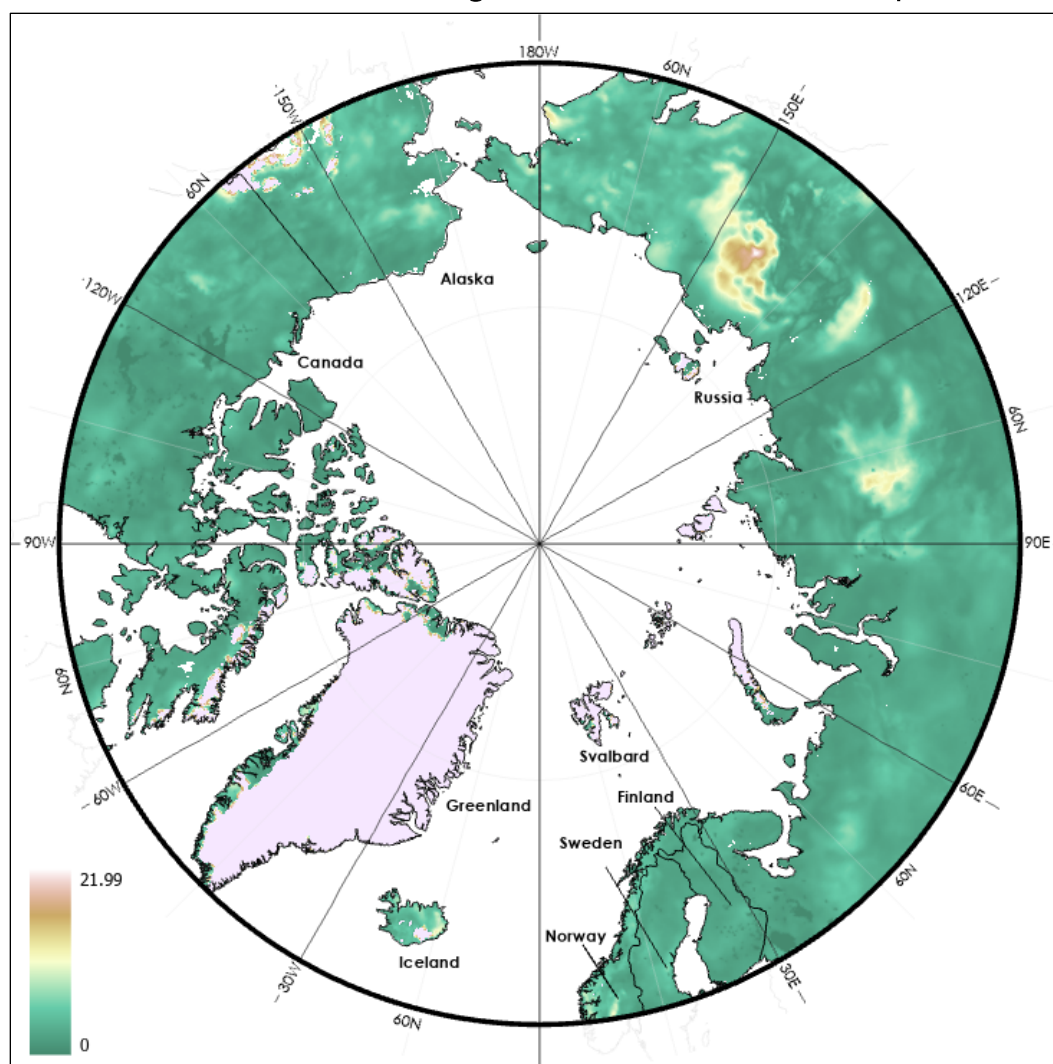


Figure A-9. The difference between the highest wind speed value (knots) and the 95th percentile wind speed, calculated based on daily averaged data computed from hourly 1991–2020 ERA5 data and then averaged across the whole October–March period.

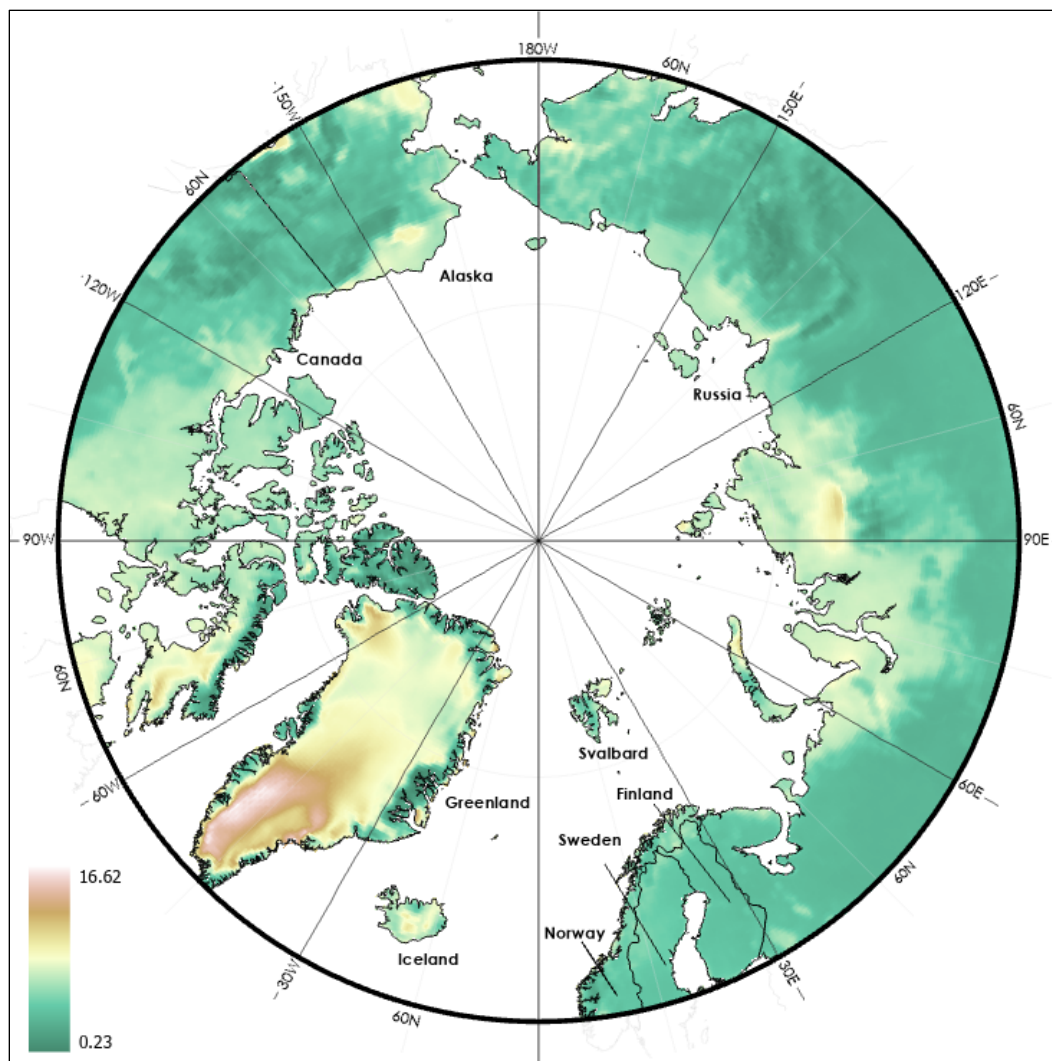


Figure A-10. The difference between the lowest windchill temperature (Fahrenheit) and the 5th percentile windchill temperature, calculated based on daily averaged data computed from hourly 1991–2020 ERA5 data and then averaged across the whole October–March period.

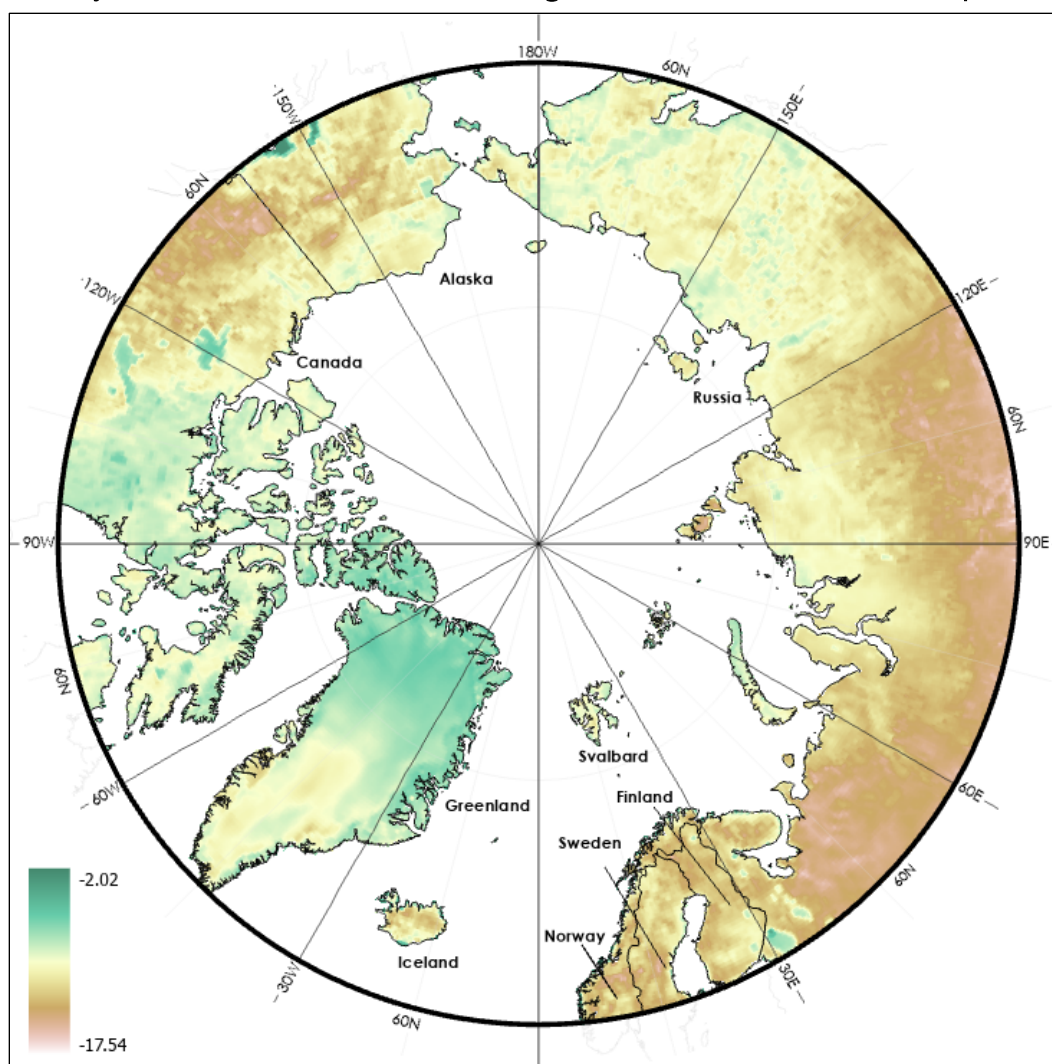
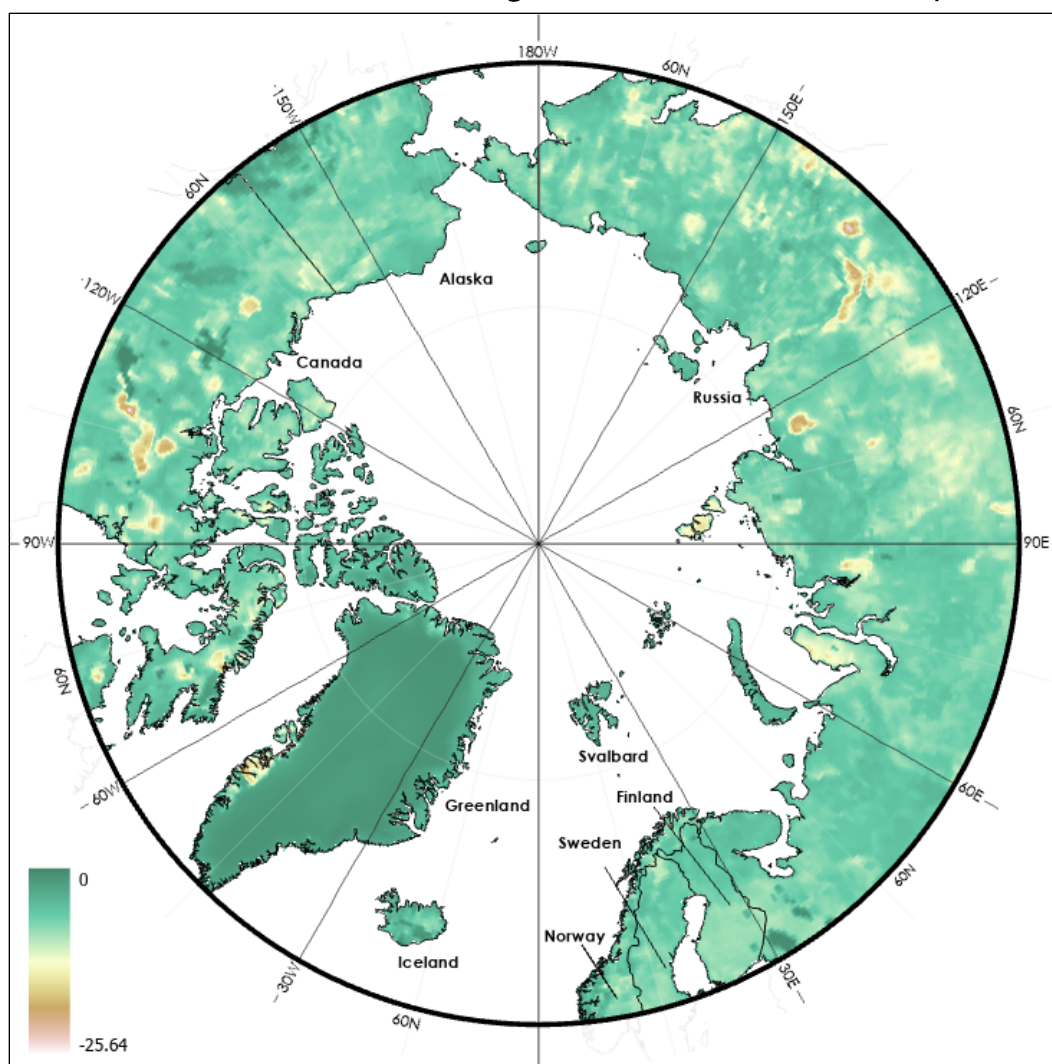


Figure A-11. The difference between the lowest soil temperature (Fahrenheit) and the 5th percentile soil temperature, calculated based on daily averaged data computed from hourly 1991–2020 ERA5 data and then averaged across the whole October–March period.



Abbreviations

ACRC	Alaska Climate Research Center
API	Application programming interface
AZCOT	Arctic Zonal Characterization and Operational Thresholding
CRREL	Cold Regions Research and Engineering Laboratory
CSV	Comma-separated value
ECMWF	European Center for Medium-Range Weather Forecasting
ERA5	European Center for Medium-Range Weather Forecasting Reanalysis v5
ERDC	Engineer Research and Development Center
FT	Frostbite time
GIS	Geographic information system
GRIB	Gridded binary
HQDA	Headquarters, Department of the Army
ICAO	International Civil Aviation Organization
MAE	Mean-absolute error
METAR	Meteorological Aerodrome Report
MSE	Mean-square error
NCAR	National Center for Atmospheric Research
NCEP	National Centers for Environmental Prediction
NCL	National Center for Atmospheric Research Command Language
netCDF	Network Common Data Format

NWP	Numerical weather prediction
PII	Personally identifiable information
POC	Point of contact
RMSE	Root-mean-square error
SD	Snow depth
SKT	Skin temperature
SL	Snow load
STD	Standard deviation
STL1	Soil temperature level 1
SWE	Snow water equivalent
10U	10 m wind velocity in the eastward direction
10V	10 m wind velocity in the northward direction
2T	2 m temperature
UI	User interface
WCT	Wind-chill temperature
WDIR	Wind direction
WD10	10 m wind direction
WGS	World Geodetic System
WSPD	Wind speed
WS10	10 m wind speed

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